The Evaporating Universe I: A dissipation principle — Resolving cosmic tensions without additional free parameters

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ABSTRACT

The ΛCDM model faces two persistent tensions: the Hubble constant measured locally ($H_0 = 73.17 \pm 0.86$ km s^{-1} Mpc^{-1}, SH0ES) disagrees at 5.8σ with the CMB inference (67.36 ± 0.54), and weak-lensing surveys measure S_8 values $\sim 2.6\sigma$ below the Planck prediction. We present the Evaporating Universe (EU), an interacting dark energy model in which a Gribov–Zwanziger ghost condensate mediates a continuous CDM → vacuum energy transfer: $Q = -\lambda\,\varepsilon(z)\,H\,\rho_c$. The dissipation fraction ($\lambda = 2/3$), logistic exponent ($b = 19/36$), transition redshift ($z_{\rm trans} \approx 5.986$), bare amplitude ($\varepsilon_{\rm bare} \approx 0.0543$), and effective coupling ($\varepsilon_{\rm IR} \approx 0.0426$) are derived from Effective Field Theory in curved spacetime; no parameter is fitted to data ($\Delta k = 0$). The model is implemented in a modified CLASS Boltzmann solver and validated through six Cobaya MCMC runs ($\sim 380{,}000$ total samples; all six converge at $R - 1 < 0.01$) against Planck CMB, DESI DR2 BAO, Pantheon+ SNe, SH0ES, and DES-Y3 weak lensing. The CDM drain yields $H_0^{\rm LKI} = 72.71 \pm 0.30$ km s^{-1} Mpc^{-1}, within 0.52σ of SH0ES. The S_8 tension is traced to a systematic bias in the ΛCDM lensing kernel; forward-modeling the EU kernel reduces the DES-Y3 discrepancy from $\sim 2.6\sigma$ to 1.03σ. Including DES-Y3 data directly in the MCMC confirms compatibility at 0.58σ. A 1024^3-particle Gadget-4 N-body simulation validates the drain signature in $P(k)$, and model selection yields $\Delta{\rm AIC} = -6.88$. We present ten zero-parameter predictions for Euclid DR1, LSST Y1, and DESI DR3, one of which — the CPL equation-of-state mirage ($w_0 = -0.855$ with CMB prior) — already matches the DESI DR2 measurement ($w_0 = -0.838 \pm 0.055$) at 0.3σ. The EU resolves both cosmological tensions simultaneously, with zero additional free parameters.

1. Introduction

Two persistent discrepancies challenge the standard ΛCDM cosmology. The Hubble constant inferred from the cosmic microwave background (CMB) under ΛCDM [46], $H_0 = 67.36 \pm 0.54$ km s^{-1} Mpc^{-1}, disagrees at 5.8σ with the local distance-ladder measurement of Breuval, Riess et al. [7], $H_0 = 73.17 \pm 0.86$ km s^{-1} Mpc^{-1}. Independently, the amplitude of matter fluctuations $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ measured by weak-lensing surveys [1] falls below the Planck ΛCDM prediction at $\sim 2.6\sigma$. These tensions persist across instruments, calibration methods, and analysis pipelines, suggesting a systematic origin rather than a statistical fluctuation.

Numerous extensions of ΛCDM have been proposed, including early dark energy [48], interacting dark energy [20], sign-switching models [2], and the CPL parametrization [12, 39] recently favored by DESI DR2 [16]. These approaches typically improve one tension at the cost of the other, or introduce free parameters whose values must be fitted to data. No published model resolves both tensions simultaneously with zero additional free parameters.

This paper presents the Evaporating Universe (EU), an IDE model in which a Gribov–Zwanziger (GZ) ghost condensate [29, 57] mediates a continuous transfer of cold dark matter (CDM) energy into the vacuum: $Q = -\lambda\,\varepsilon(z)\,H\,\rho_c$, where $\lambda = 2/3$ is the transversality fraction on the spatial hypersurface and $\varepsilon(z)$ is a logistic activation function whose parameters are determined by the heat-kernel threshold, the CKN holographic bound [14], and the TT projection of the ghost–graviton vertex. All five model parameters are derived from Effective Field Theory in curved spacetime; none are fitted to observational data ($\Delta k = 0$).

The CDM drain modifies the Friedmann equation at the percent level, producing two observable effects. The *global kinematic imprint* (GKI) raises the background expansion rate to $H_0^{\rm GKI} = 68.89$ km s^{-1} Mpc^{-1}. The *local kinematic imprint* (LKI), arising from the KBC void outflow [34, 62], boosts the locally measured value to $H_0^{\rm LKI} = 72.71$ km s^{-1} Mpc^{-1} — within 0.52σ of SH0ES. The growth tension is resolved by identifying a systematic bias in the weak-lensing pipeline: the ΛCDM lensing kernel applied to EU-generated data underestimates S_8 by construction. Forward-modeling the EU kernel reduces the DES-Y3 tension from $\sim 2.6\sigma$ to 1.03σ; including DES-Y3 data directly in the MCMC confirms compatibility at 0.58σ. Model selection — comparing both models on identical datasets with the KBC void correction applied to both — yields $\Delta{\rm AIC} = -6.88$ favoring the EU.

The paper is organized as follows. Section 2 derives the five EU parameters from the GZ framework. Section 3 presents the analytic predictions for H_0. Section 4 describes the CLASS-EU implementation and stability properties. Section 5 reports the MCMC analysis across six convergence runs. Section 6 characterizes the drain signature in cosmological observables. Section 7 resolves the Hubble tension via the GKI/LKI decomposition. Section 8 validates the drain signature with N-body simulations. Section 9

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resolves the growth tension through lensing kernel forensics. Section 10 presents ten falsifiable predictions for Euclid, LSST, and DESI. Section 11 places the EU in the landscape of beyond-ΛCDM models, discusses theoretical limitations, and outlines observational tests. Section 12 summarizes the results.

2. Theoretical Framework

We derive the five parameters of the Evaporating Universe from Effective Field Theory in curved spacetime, utilizing the topological restrictions of the Gribov–Zwanziger framework and holographic bounds. No parameter is fitted to observational data ($\Delta k = 0$).

2.1. The Gribov–Zwanziger ghost condensate

The Gribov–Zwanziger (GZ) framework addresses a fundamental obstacle in the quantization of non-Abelian gauge theories: the existence of gauge-equivalent configurations (Gribov copies) that survive the Landau gauge condition $\partial_\mu A^a_\mu = 0$ [29]. The resolution restricts the functional integral to the first Gribov region,

$$\Omega = \left\{ A^a_\mu \;\middle|\; \partial_\mu A^a_\mu = 0,\ \ \mathcal{M}^{ab} > 0 \right\}, \tag{1}$$

where $\mathcal{M}^{ab} = -\partial_\mu D^{ab}_\mu$ is the Faddeev–Popov operator [57]. This restriction generates a non-perturbative ghost condensate $\langle \bar{c}^a c^a \rangle \neq 0$ — a vacuum expectation value absent in perturbation theory.

Through Ashtekar's reformulation of general relativity as an SU(2) gauge theory on the spatial slice Σ [5], the GZ framework extends to the gravitational sector [11]. The ghost condensate on the Friedmann–Lemaître–Robertson–Walker (FLRW) background acts as the order parameter for a cosmological phase transition: below a critical redshift $z_{\rm trans} \approx 6$, the condensate activates a continuous transfer of energy from cold dark matter (CDM) to the vacuum.

CDM participates in this transfer because it is a gauge singlet: carrying no Standard Model charge, it interacts only gravitationally and couples to the bulk vacuum through the ghost condensate. Baryonic matter, bound to Σ by its $\mathrm{SU(3)_{QCD}} \times \mathrm{U(1)_{EM}}$ gauge charges, is protected from evaporation by those charges.

2.2. The unique CDM–vacuum interaction

The modified continuity equations for CDM, vacuum energy, and baryons are

$$\dot{\rho_{\rm c}} + 3H\rho_{\rm c} = -Q, \tag{2}$$
$$\dot{\rho_\Lambda} = +Q, \tag{3}$$
$$\dot{\rho_{\rm b}} + 3H\rho_{\rm b} = 0, \tag{4}$$

where $Q \geq 0$ is the interaction rate. Total energy conservation is guaranteed by construction: Eqs. (2)–(4) sum to the Bianchi identity $\dot{\rho}_{\rm tot} + 3H(\rho_{\rm tot} + p_{\rm tot}) = 0$.

The functional form of Q is the unique leading-order interaction consistent with FLRW symmetry and decay kinematics.

(i) Dimension. From Eq. (2), $[Q] = [\rho]\,[T^{-1}]$: an energy density per unit time.

(ii) Initial-state population. The interaction describes CDM → vacuum transfer. By Fermi's golden rule, the rate is proportional to the initial-state population: $Q \propto \rho_{\rm c}$. This excludes $Q \propto \rho_\Lambda$ and mixed forms.

(iii) Geometric clock. The factor $[T^{-1}]$ must be a homogeneous, isotropic, first-order geometric scalar. In FLRW spacetime, the unique candidate is $H = \dot{a}/a$. Higher-order terms ($\dot{H}/H,\ H^2/H_0$) are sub-leading in the derivative expansion; particle-physics scales break geometric universality; the cosmic time $1/t$ is not covariant under time reparametrization.

The unique leading-order interaction is therefore

$$\boxed{Q = \lambda\,\varepsilon(z)\,H\,\rho_{\rm c}\,,} \tag{5}$$

where λ is the dissipation fraction and $\varepsilon(z)$ is the redshift-dependent coupling amplitude, both derived in Sec. 2.3. The interacting dark energy (IDE) literature [56, 25, 20] postulates this form; Eq. (5) establishes it as the unique solution of the derivative expansion.

The coupling amplitude $\varepsilon(z)$ is the order parameter of the ghost condensate. Below the condensation threshold $z_{\rm trans}$, it follows a generalized logistic profile with a strict Heaviside cutoff (Fig. 2(a)):

$$\varepsilon(z) = \begin{cases} \dfrac{\varepsilon_{\rm IR}}{1 + \left(\dfrac{1+z}{1+z_{\rm trans}}\right)^{1/b}} & z \leq z_{\rm trans}, \\ 0 & z > z_{\rm trans}. \end{cases} \tag{6}$$

The cutoff encodes the physics of the condensation: the ghost order parameter is strictly zero above the threshold, so no CDM is drained at $z > z_{\rm trans}$. The CMB, big bang nucleosynthesis, and all pre-condensation physics are exactly ΛCDM. At late times ($z \to 0$), $\varepsilon \to \varepsilon_{\rm IR}$ (full coupling); at the transition ($z = z_{\rm trans}$), $\varepsilon = \varepsilon_{\rm IR}/2$ (onset).

2.3. Framework-derived parameters

The five quantities $\{\lambda,\ b,\ z_{\rm trans},\ \varepsilon_{\rm bare},\ \varepsilon_{\rm IR}\}$ entering Eqs. (5)–(6) are each determined by the mathematical structure of the GZ framework. We derive them in order of increasing complexity.

2.3.1. *Dissipation fraction:* $\lambda = 2/3$

The dissipation fraction quantifies the fraction of ghost-field channels available for CDM–vacuum energy transfer. Its value follows from the transversality structure of the ghost field on the spatial slice Σ.

Under the ADM 3+1 decomposition, the ghost field $c^\mu = (c^0, c^i)$ separates into a constrained temporal component c^0 and three spatial components c^i on Σ. The Helmholtz decomposition theorem yields

$$c^i = c^i_T + \partial^i \varphi, \qquad \partial_i c^i_T = 0, \tag{7}$$

where c_T^i carries two transverse degrees of freedom and φ is a longitudinal scalar.

Two independent arguments force $\varphi = 0$. First, the Landau gauge condition $\partial_\mu A_\mu^a = 0$ projects the Faddeev–Popov operator onto the transverse functional space, requiring $\partial_i c^i = 0$ and hence $\nabla^2\varphi = 0$ [29]. Second, a longitudinal condensate $\langle \bar{c}_\parallel^a c_\parallel^b \rangle \neq 0$ would single out a preferred direction, spontaneously breaking the isotropy of the FLRW background. Both routes eliminate the longitudinal mode: only the two transverse polarizations propagate within Ω.

The GZ effective action confirms this counting. At one-loop order in d dimensions [57],

$$\Gamma_\gamma^{(1)} \supset \underbrace{-d\,(N^2-1)\,\gamma^4}_{\text{tree: } d \text{ modes}} + \underbrace{\frac{N^2-1}{2}(d-1)\int \frac{d^d q}{(2\pi)^d} \ln \frac{q^4 + 2g^2 N\gamma^4}{q^2}}_{\text{loop: } (d-1) \text{ transverse modes}}\,, \tag{8}$$

where the tree-level prefactor d sums over all Lorentz indices and the one-loop prefactor $(d-1)$ reflects the Landau gauge projection. The gap equation $\partial\Gamma/\partial\gamma^2 = 0$ produces the ratio $(d-1)/d$ as an algebraic quotient of two fixed coefficients in the action. On the spatial section of FLRW ($d \to n = 3$):

$$\boxed{\lambda = \frac{n-1}{n} = \frac{2}{3}\,.} \tag{9}$$

The stability of this result under radiative corrections follows from the Slavnov–Taylor identity [53] for the ghost–gluon vertex in the Landau gauge:

$$Z_c = Z_g^{-1}\, Z_A^{-1/2}\,, \tag{10}$$

where Z_c, Z_g, and Z_A are the renormalization constants of the ghost field, coupling, and gluon field, respectively. This identity constrains the ghost self-energy to all orders: the transverse fraction $(n-1)/n$ entering the gap equation is not an artifact of the one-loop truncation but a consequence of the gauge structure itself. Dudal et al. [22] verified that the refined Gribov–Zwanziger action preserves this identity, so the channel counting and the gap equation are anchored to the same algebraic structure. Any higher-loop correction that modified λ would violate Eq. (10).

2.3.2. *Logistic exponent:* $b = 19/36$

The exponent b in Eq. (6) controls the sharpness of the condensation transition. It equals the fraction of the ghost-graviton interaction that couples to the physical (transverse-traceless) graviton polarizations in $d = 4$ spacetime.

The ghost-graviton vertex $V^\mu_{\nu;\rho\sigma}(p,q)$ from linearized quantum gravity [13] couples the Faddeev–Popov ghost to the symmetric graviton tensor. The physical content is extracted by projecting onto the transverse-traceless (TT) sector via the York projector $\Pi^{\rm TT}_{\rho\sigma,\lambda\tau}(k)$ [63]. The TT fraction is

$$\boxed{b = \frac{\mathrm{Tr}\big(V\cdot\Pi^{\rm TT}\cdot V\big)}{\mathrm{Tr}(V\cdot\mathbb{I}\cdot V)} = \frac{19}{36}\,,} \tag{11}$$

verified with irreducible exact rational arithmetic across four independent momentum configurations (Appendix A.1). The numerator 19 is prime, confirming irreducibility — a genuine $d = 4$ tensor invariant.

The TT fraction governs the logistic activation through the renormalization group (RG). Only the TT-projected component of the vertex drives the condensate running — analogous to active quark flavors modifying the QCD β-function — so the ghost self-energy acquires the anomalous dimension b, yielding the Wilson critical exponent $\nu = 1/b = 36/19$. The logistic β-function,

$$\frac{d\sigma}{d\ln\mu} = -\frac{1}{b}\,\sigma\left(1 - \frac{\sigma}{\sigma_*}\right), \tag{12}$$

is exact at the upper critical dimension $d = 4$: higher-order corrections are strictly irrelevant by the Ginzburg criterion, protecting the universality of the logistic form. The RG scale maps to redshift through the CKN saturation (Sec. 2.3.3): $\Lambda_{\rm UV}^4 = \rho_{\rm crit} \propto H^2$ implies $\mu \propto H^{1/2}$, a monotonically increasing function of $(1+z)$. The logistic solution depends only on the ratio $(1+z)/(1+z_{\rm trans})$, rendering the result robust against the precise functional form of $\mu(z)$.

2.3.3. *Condensation threshold:* $z_{\rm trans} \approx 5.986$

The transition redshift marks the onset of the ghost condensation. It is determined by a closed chain of five links, each independently established.

Link 1: One-loop amplitude. The universal one-loop prefactor from the Seeley–DeWitt heat-kernel expansion [58] in $d = 4$ dimensions is

$$(4\pi)^{-d/2}\Big|_{d=4} = \frac{1}{16\pi^2} \approx 0.00633\,. \tag{13}$$

This is the standard phase-space suppression of a single quantum loop, appearing identically in dimensional regularization, the Coleman–Weinberg potential, and the ABJ anomaly.

Link 2: Holographic bound. Cohen, Kaplan, and Nelson [14] established that an effective field theory in a region of size L must satisfy $L^3\Lambda_{\rm UV}^4 \lesssim LM_{\rm Pl}^2$. Applied to the Hubble sphere ($L = H^{-1}$) with the exact Schwarzschild condition:

$$\Lambda_{\rm UV}^4 \leq \frac{3M_{\rm Pl}^2 H^2}{8\pi} = \rho_{\rm crit}\,. \tag{14}$$

Link 3: Saturation. The CKN bound (Eq. 14) is formally an inequality. In a spatially flat universe ($\Omega_k = 0.001 \pm 0.002$ [46]), the late-time de Sitter attractor drives $\Omega_\Lambda \to 1$, saturating the Gibbons–Hawking entropy bound $S = A/(4G)$ as a thermodynamic identity [27]. The CKN energy bound is the effective field theory formulation of this holographic constraint [14]; the inequality becomes an equality: $\Lambda_{\rm UV}^4 = \rho_{\rm crit}$. Conversely, exact saturation requires $\Omega_k = 0$: any spatial curvature introduces a time-dependent correction $\propto k/(a^2H^2)$, breaking the constancy of the bound. The observed flatness and CKN saturation are mutually required.

Link 4: Threshold. With $\Lambda_{\rm UV}^4 = \rho_{\rm crit}$ (Link 3), the cosmological density parameter $\Omega_{\rm DE} \equiv \rho_\Lambda/\rho_{\rm crit}$ measures the vacuum energy as a fraction of the total UV phase space. Combining with Link 1, the condensate nucleates when this fraction reaches the one-loop amplitude:

$$\Omega_{\rm DE}(z_{\rm trans}) = \frac{1}{16\pi^2}\,. \tag{15}$$

Link 5: Solution. In the matter-dominated ΛCDM background (valid at $z \sim 6$ where EU and radiation corrections are negligible):

$$\boxed{z_{\rm trans} = \left[\frac{\Omega_\Lambda\,(16\pi^2-1)}{\Omega_{\rm m}}\right]^{1/3} - 1 \approx 5.986\,.} \tag{16}$$

The inputs are $\Omega_\Lambda = 0.6847$ and $\Omega_{\rm m} = 0.3153$ from Planck 2018 [46]. Because $\varepsilon(z) = 0$ for $z > z_{\rm trans}$ (Eq. 6), the pre-condensation background is rigorously ΛCDM; Eq. (16) therefore evaluates exactly using the primordial densities. The sensitivity to $h = H_0/100$ is weak: $z_{\rm trans}$ varies by $\sim 6\%$ over $h \in [0.66, 0.70]$ (Appendix A.2).

2.3.4. Coupling amplitude: $\varepsilon_{\rm bare}$ and $\varepsilon_{\rm IR}$

The coupling amplitude has two components: a bare value set by the vacuum channel counting and an infrared screening from the Gribov horizon.

Bare coupling. The Gribov–Zwanziger framework applies to any gauge theory formulated in terms of a connection. General relativity on Σ, as conventionally written in metric variables, is not a gauge theory in this sense. Ashtekar's canonical reformulation [5] resolves this by recasting the gravitational phase space in terms of an SU(2) connection A_a^i and a densitized triad $\tilde{E}_i^a$ on the spatial slice. Einstein's constraints become polynomial in these variables, and the internal gauge group is SU(2) with $\dim(\mathrm{SU}(2)) = n = 3$. The existence of Gribov copies is a topological property of all non-Abelian gauge theories in the Landau gauge, independent of the specific gauge group [57]. This SU(2) identification therefore places the gravitational sector on the same formal footing as QCD, making the Faddeev–Popov operator, the Gribov region Ω, and its boundary ∂Ω well-defined.

Within this gauge-theoretic formulation, the GZ localizing field φ_a^{ij} [65] — an auxiliary tensor coupling the ghost (c^i) and anti-ghost ($\bar{c}^j$) to the gauge index a — possesses three independent indices, each running from 1 to n. In the deep UV, where the ghost and anti-ghost are distinct fermionic fields with unrestricted quantum numbers (Zwanziger's independent-index theorem [65]), the auxiliary field fluctuates as a fully general rank-3 tensor: $N_{\rm bare} = n^3 = 27$ bare channels. In the classical IR, the metric symmetry $\gamma_{ij} = \gamma_{ji}$ constrains the ghost indices, reducing the auxiliary field to $n\,n(n+1)/2$ independent components: $N_{\rm classical} = 18$ for $n = 3$.

The Gribov projection connects these two regimes. Setting the effective count $\lambda\,n^3 = n^2(n-1)$ equal to the classical count:

$$n^2(n-1) = \frac{n^2(n+1)}{2} \quad \Longrightarrow \quad \boxed{n = 3\,.} \tag{17}$$

The quantum vacuum recovers the classical limit only in three spatial dimensions: $(2/3)(27) = 18 = 3(3+1)/2 \cdot 3$. This consistency condition is not imposed; it is a consequence of requiring the GZ ghost sector to reproduce the correct number of gravitational degrees of freedom.

Democratic coupling. CDM carries no Standard Model gauge charge and, crucially, no internal SU(2) quantum numbers under the Ashtekar connection. It is a gauge singlet. By the Wigner–Eckart theorem, the matrix element of any gauge-invariant operator between singlet states is proportional to the identity on the mediator space: no channel is preferred over any other. CDM therefore couples with equal amplitude to all 27 bare channels. The transverse one-loop factor $(4\pi)^{-1}$ — reflecting the restriction of the ghost dynamics to the two-dimensional transverse plane on Σ (Sec. 2.3.1), in contrast to the bulk factor $(4\pi)^{-2}$ used for the condensation threshold — distributes uniformly over the bare modes, yielding the phase-space fraction $1/(4\pi\,n^3)$. By Fermi's golden rule, the transition rate scales as $|\varepsilon|^2$; the coupling amplitude is therefore the square root of this fraction:

$$\boxed{\varepsilon_{\rm bare} = \frac{1}{\sqrt{4\pi\,n^3}} = \frac{1}{\sqrt{108\pi}} \approx 0.05429\,.} \tag{18}$$

IR screening. In the deep infrared, the Gribov horizon modifies the ghost propagator from $D_{\rm UV}(k) = 1/(k^2+\gamma^2)$ to $D_{\rm GZ}(k) = k^2/(k^4+\gamma^4)$. This deformation of the internal propagator acts as a vertex renormalization, screening the bare coupling by a calculable factor. The ratio of integrated propagator densities on the two-dimensional transverse momentum plane [66] evaluates exactly to (Appendix A.3)

$$\frac{I_{\rm IR}}{I_{\rm UV}} = \frac{\pi}{4}\,. \tag{19}$$

The effective coupling is therefore

$$\boxed{\varepsilon_{\rm IR} = \frac{\pi}{4}\,\varepsilon_{\rm bare} \approx 0.04264\,.} \tag{20}$$

Table 1
The five EU parameters. All are derived from the mathematical structure of the Gribov–Zwanziger EFT in curved spacetime. None are fitted to observational data ($\Delta k = 0$).

Parameter	Value	Exact form	Origin
λ	$0.6\bar{6}$	2/3	GZ transversality
b	0.5278	19/36	TT vertex fraction
$z_{\rm trans}$	5.986	Eq. (16)	Heat kernel + CKN
$\varepsilon_{\rm bare}$	0.05429	$1/\sqrt{108\pi}$	GZ channels
$\varepsilon_{\rm IR}$	0.04264	$(\pi/4)\,\varepsilon_{\rm bare}$	Gribov screening

2.4. Modified Friedmann equation

The CDM density evolves under the combined effect of Hubble dilution and vacuum transfer. Integrating Eq. (2):

$$\rho_c(a) = \rho_{c,0}\, a^{-3}\, f_{\rm surv}(a), \tag{21}$$

where the survival fraction is

$$f_{\rm surv}(a) = \exp\left[-\lambda \int_0^a \varepsilon(a')\,\frac{da'}{a'}\right]. \tag{22}$$

The Heaviside cutoff in Eq. (6) renders the total accumulated drain analytically tractable. The kinematic integral is

$$\mathcal{I} = \varepsilon_{\rm IR}\, b\left[\ln\!\left(1 + (1{+}z_{\rm trans})^{1/b}\right) - \ln 2\right] \approx 0.0679\,, \tag{23}$$

where the $-\ln 2$ arises from the upper limit of integration at $z_{\rm trans}$. The CDM survival fraction at $z = 0$ is

$$f_{\rm surv}(0) = e^{-\lambda \mathcal{I}} \approx 0.9558\,, \tag{24}$$

corresponding to a 4.4% transfer of CDM energy to the vacuum since $z \approx 6$.

The modified Friedmann equation is

$$H^2(a) = H_0^2\big[\Omega_{\rm r}\, a^{-4} + \Omega_{\rm b}\, a^{-3} + \Omega_{\rm c}\, a^{-3}\, f_{\rm surv}(a) + \Omega_\Lambda(a)\big], \tag{25}$$

where $\Omega_{\rm r}$ encodes photons and relativistic neutrinos (one massive species with $\sum m_\nu = 0.06$ eV [46]), $\Omega_{\rm c}$ denotes the pre-drain CDM density parameter anchored by the CMB ($\varepsilon = 0$ at $z > z_{\rm trans}$, so $\Omega_{\rm c}$ retains its ΛCDM value), and $\Omega_\Lambda(a)$ is obtained by integrating Eq. (3) numerically. The CDM and vacuum terms do not cancel: drained CDM energy acquires the vacuum equation of state ($w = -1$) and ceases to dilute, generating a net modification to $H(z)$. The full numerical integration is performed by the CLASS Boltzmann solver [6] (Sec. 4).

Despite absorbing CDM energy, the vacuum retains $w_\Lambda = -1$ exactly — a property proven algebraically in Sec. 4.2.

With all five parameters determined, the model predicts two distinct kinematic imprints on the Hubble rate: a global shift from CDM drain and a local boost from the KBC void outflow.

3. Kinematic Imprints: GKI and LKI

The five framework-derived parameters produce two distinct kinematic signatures on the Hubble rate. The *Global Kinematic Imprint* (GKI) modifies the background expansion through CDM drain, while the *Local Kinematic Imprint* (LKI) biases local distance-ladder measurements through the KBC void outflow. Both are zero-parameter predictions.

3.1. Global Kinematic Imprint

The CMB power spectrum fixes the physical CDM density $\omega_{\rm cdm} \equiv \Omega_{\rm c} h^2 = 0.1200 \pm 0.0012$ [46] at $z \sim 1100$, where the EU coupling is strictly zero ($\varepsilon = 0$ for $z > z_{\rm trans}$). The CMB therefore measures the same $\omega_{\rm cdm}$ as ΛCDM.

If CDM evaporates at late times, $\Omega_{\rm c}$ decreases while $\omega_{\rm cdm}$ remains CMB-anchored. The reduced Hubble parameter h must increase to maintain the constraint:

$$h^{\rm EU} = \frac{h^{\rm Planck}}{\sqrt{f_{\rm surv}(0)}}\,, \tag{26}$$

yielding the global prediction

$$\boxed{H_0^{\rm GKI} = \frac{H_0^{\rm Planck}}{\sqrt{f_{\rm surv}(0)}} = \frac{67.36}{\sqrt{0.9558}} \approx 68.90\ {\rm km\,s^{-1}\,Mpc^{-1}}\,.} \tag{27}$$

Because CDM has evaporated, the present-day physical matter density is $\omega_m(z{=}0) = \omega_b + \omega_{\rm cdm}\, f_{\rm surv}(0) \approx 0.137$. The corresponding matter density parameter is $\Omega_m^{\rm EU} = \omega_m(z{=}0)/(h^{\rm EU})^2 \approx 0.289$, compared with $\Omega_m^{\Lambda{\rm CDM}} = 0.3153$.

This mechanism — CDM drain reducing $\Omega_{\rm c}$ and raising h — is the same physics exploited by Interacting Dark Energy models [20], but with a coupling fixed by the framework ($\lambda = 2/3$) rather than fitted to data.

3.2. Local Kinematic Imprint

The GKI predicts the *global* expansion rate. Local distance-ladder measurements are systematically higher because the observer sits inside the KBC void [34], a local underdensity extending to $R \sim 300$ Mpc with observed contrast $\delta_{\rm obs} = -0.46 \pm 0.06$.

The void outflow biases the locally inferred H_0 (Wu & Huterer 2017 [62]):

$$\frac{\delta H_0}{H_0} = -\frac{1}{3}\,\delta_{\rm true}\, f(\Omega_m)\,\Theta\,, \tag{28}$$

where $f(\Omega_m)$ is the linear growth rate and Θ accounts for the non-linear correction of Marra et al. [43].

Two refinements enter the calculation. First, the observed contrast $\delta_{\rm obs}$ is inflated by redshift-space distortions (Haslbauer, Banik & Kroupa 2020 [31]):

$$\delta_{\rm true} = \frac{\delta_{\rm obs}}{1 + f/b_g} = \frac{-0.46}{1 + 0.514/1.0} \approx -0.304\,, \tag{29}$$

where $f = 0.514$ is computed from the exact Riccati ODE [40] evaluated in the EU background (not the $\Omega_m^{0.55}$ approximation) and $b_g = 1.0$ is the K-band galaxy bias. Second, the non-linear correction $\Theta = 1.064$ amplifies the outflow by 6.4% relative to linear theory [43].

The local prediction is

$$\begin{aligned} H_0^{\rm LKI} &= H_0^{\rm GKI} + \delta H_0 \\ &= 68.90 + 3.82 \approx \boxed{72.72\ {\rm km\,s^{-1}\,Mpc^{-1}}\,.} \end{aligned} \tag{30}$$

CDM drain also causes adiabatic halo expansion [28], but its effect on the distance ladder is negligible: $\Delta R_{\rm halo} \sim 10$ kpc versus $D_{\rm SN} \sim 40$ Mpc implies $\Delta H_0 \sim 0.009$ km s^{-1} Mpc^{-1}.

Sensitivity. The LKI prediction is robust against the KBC void uncertainty. Propagating $\delta_{\rm obs} = -0.46 \pm 0.06$ through the full RSD-corrected formula yields $H_0^{\rm LKI} \in [72.19, 73.26]$ at 1σ, with worst-case tension against SH0ES of 1.14σ.

3.3. Observational hierarchy

The GKI and LKI channels predict a characteristic hierarchy: global probes recover H_0 consistent with the Planck value, while local calibrators — Cepheids, TRGB, JAGB — converge near $H_0^{\rm LKI} \approx 72.7$, as their Rung 3 supernovae lie within the KBC void.

Table 2
EU predictions for the Hubble constant. The CMB prediction is the GKI (global); all local tracers receive the same LKI (void) correction. TRGB/JAGB values from Freedman et al. [24]. All values in km s^{-1} Mpc^{-1}.

Tracer	$H_0^{\rm obs}$	σ	$H_0^{\rm EU}$	Tension
CMB/Planck	67.36	0.54	67.36	0.0σ
JAGB/JWST	67.80	2.72	72.72	1.8σ
TRGB/JWST	68.81	2.22	72.72	1.8σ
TRGB (mixed)	70.39	1.94	72.72	1.2σ
SH0ES 2024	73.17	0.86	72.72	0.5σ

Table 2 summarizes the comparison. All five tracers are consistent with the EU prediction at $< 2\sigma$. The analytical predictions of this section are validated by the full Boltzmann integration in Sec. 4.

The same CDM drain that modifies H_0 also suppresses structure growth by reducing Ω_m and altering $H(z)$. The quantitative resolution of the S_8 tension is presented in Sec. 9.

4. Numerical Implementation

We implement the EU cosmology in the Cosmic Linear Anisotropy Solving System (CLASS v3.3.4 [6]), verifying that the analytic predictions of Sec. 3 are reproduced by the full Boltzmann integration. The implementation modifies four source files; the complete patched source, analysis notebooks, and MCMC chains are publicly available [4].

4.1. Solver architecture

Four patches are applied to the CLASS source. (i) The header `background.h` receives the five EU parameters ($\varepsilon_{\rm IR}$, $z_{\rm trans}$, b, λ, and the perturbation switch). (ii) The parser `input.c` reads these parameters from the configuration file, defaulting to $\varepsilon_{\rm IR} = 0$ (standard ΛCDM). (iii) The background integrator `background.c` implements the coupled ODE system

$$\frac{d f_{\rm cdm}}{d \ln a} = -\lambda\, \varepsilon(z)\, f_{\rm cdm}\,, \tag{31}$$

$$\frac{d f_{\rm vac}}{d \ln a} = +\lambda\, \varepsilon(z)\, f_{\rm cdm}\, a^{-3}\,, \tag{32}$$

where both $f_{\rm cdm}$ and $f_{\rm vac}$ are normalized to $\rho_{c,0} \equiv \Omega_{c,0}\, H_0^2/(8\pi G/3)$. The system is solved by a fourth-order Runge–Kutta integrator over 2000 steps in $\ln a \in [-7, 0]$, with initial conditions $f_{\rm cdm} = 1$, $f_{\rm vac} = 0$. The CDM density is multiplied by $f_{\rm cdm}(a)$ at each background evaluation, and the vacuum receives the Friedmann closure injection $\rho_{c,0}\, f_{\rm vac}(a)$. (iv) The perturbation integrator `perturbations.c` receives the interacting dark energy (IDE) source terms of Valiviita, Majerotto & Maartens (2008) [56].

The EU and ΛCDM runs share identical cosmological parameters (Planck 2018 baseline [46]). The ΛCDM run fixes $H_0 = 67.36$ km s^{-1} Mpc^{-1}; the EU run inputs the same $100\theta_s$ and lets CLASS shoot for H_0, which is the standard procedure when comparing models with different late-time physics.

4.2. Perturbation stability

The EU coupling $Q = -\lambda\, \varepsilon(z)\, H\, \rho_{\rm c}$ is proportional to the local CDM density. This proportionality guarantees algebraic cancellation of the IDE perturbation source terms for $w_\Lambda = -1$:

Continuity. The source term in the CDM overdensity equation is proportional to $[\delta Q/Q - \delta_c]$. Since $Q \propto \rho_{\rm c}$, the fractional perturbation $\delta Q/Q = \delta_c$, yielding $[\delta_c - \delta_c] = 0$.

Euler. The momentum drag is proportional to $[\theta_{\rm de} - \theta_{\rm cdm}]$. For $w = -1$ (vacuum), $\theta_{\rm de} = 0$; in synchronous gauge, $\theta_{\rm cdm} = 0$ by construction. Hence $[0 - 0] = 0$.

Dense and underdense regions evaporate at the same fractional rate. Structure growth is modified only through the background: the diluted Poisson source (weaker $\rho_{\rm cdm}$) produces shallower potential wells.

Null test. We verify this cancellation by running CLASS twice with identical parameters but the IDE perturbation switch toggled on and off. The difference in σ_8 is

$$\begin{aligned}\Delta\sigma_8 &= |\sigma_8^{\rm IDE\ on} - \sigma_8^{\rm IDE\ off}| \\ &= 0 \quad \text{(exact to machine precision)}\,.\end{aligned} \tag{33}$$

Doom factor. Standard IDE models with $w \neq -1$ suffer from large-scale instabilities when the ratio $Q/[(1+w)\rho_{\rm de}]$ diverges [25, 56]. In the EU, $w_\Lambda = -1$ exactly: the vacuum does not cluster ($\delta_{\rm de} = 0$) and carries no velocity perturbations ($\theta_{\rm de} = 0$). The dark energy Euler equation drops out of the perturbation system, so the singular ratio never arises. Unlike standard IDE models where w is a free parameter that must be tuned away from -1 to avoid this instability, the EU derives $w_\Lambda = -1$ algebraically from the Cancellation Theorem. The doom factor instability is thus structurally absent — a consequence of the model's internal consistency, not a numerical choice.

4.3. Background evolution

Table 3 compares the $z = 0$ background quantities from the fiducial CLASS runs (Planck 2018 input parameters). The EU CDM drain of 4.42% is confirmed, and energy conservation is exact: $|\Omega_{\rm tot} - 1| < 10^{-15}$ (Bianchi identity).

The vacuum equation of state from CLASS reads $w_\Lambda = -1.003$, consistent with $w = -1$ exact to within the RK4

Table 3
Background quantities at $z = 0$ from CLASS. Both runs use Planck 2018 input parameters; the EU run shoots for H_0 via θ_s matching.

Quantity	ΛCDM	EU	Shift
H_0 (km s^{-1} Mpc^{-1})	67.36	68.56	+1.8%
Ω_m	0.315	0.293	−7.1%
Ω_Λ	0.685	0.707	+3.2%
$f_{\rm cdm}(0)$	1.000	0.956	−4.4%
r_d (Mpc)	147.06	147.06	0.00%
σ_8	0.811	0.829	+2.3%
S_8	0.831	0.819	−1.4%

integration residual ($\sim$ 0.3%). The sound horizon r_d is identical between the two models because the EU coupling is zero at $z > z_{\rm trans} \approx 6$, leaving recombination physics unchanged. The fiducial $H_0 = 68.56$ in Table 3 reflects the background response under fixed Planck primordial densities; re-evaluation at the MCMC posteriors (Sec. 6) shifts H_0 upward into sub-percent agreement with the analytic prediction of Sec. 3.1.

4.4. CMB power spectrum

The C_ℓ^{TT} power spectra of the two models are virtually indistinguishable at $\ell > 30$, where the acoustic peaks are determined by pre-recombination physics ($z > z_{\rm trans}$). At $\ell \lesssim 20$, the EU spectrum shows a modest enhancement due to the late integrated Sachs–Wolfe (ISW) effect: CDM drain reduces the matter density at $z < 2$, causing gravitational potentials to decay faster and boosting the low-ℓ power. This $\sim$ 5% ISW enhancement is a falsifiable prediction of the model.

With the solver validated at both background and perturbation level, we proceed to the statistical analysis in Sec. 5.

The development of the CLASS-EU solver patches, Cobaya configuration files, and Gadget-4 simulation scripts was supported by an AI coding assistant used for code development, debugging, and optimization. All numerical outputs were independently verified by the author through convergence diagnostics ($R - 1 < 0.01$), analytical cross-checks (Table 5), and null tests (Table **??**).

5. Statistical Analysis

We confront the EU model with cosmological data via Markov Chain Monte Carlo (MCMC) sampling, using Cobaya [54] as the sampler and CLASS-EU (Sec. 4) as the theory engine.

5.1. Sampling strategy

The analysis comprises six MCMC runs organized in three phases across two parameter configurations.

The first configuration, *EU-blind* ($\Delta k = 3$), samples the three EU parameters ($\varepsilon_{\rm IR}$, $z_{\rm trans}$, b) with flat priors alongside the standard cosmological parameters. The second, *EU* ($\Delta k = 0$), fixes all EU parameters at the framework-derived values of Sec. 2.3.

Phase 1 runs each configuration with the IDE perturbation source terms disabled, generating stable proposal covariance matrices. *Phase 2* activates the source terms and uses the Phase 1 covariance as the proposal, producing the baseline posteriors. *Phase 3* extends the EU configuration with local measurements: *EU+H_0* adds the SH0ES Cepheid distance-ladder measurement [7]; *EU+H_0+S_8* adds DES-Y3 3 × 2pt cosmic shear via Cocoa/CosmoLike [1, 36], sampling 23 additional nuisance parameters (photometric redshifts, shear calibration, intrinsic alignments, and galaxy bias).

All six runs share a baseline likelihood consisting of Planck PR4 NPIPE CamSpec TTTEEE, low-ℓ TT and EE, CMB lensing $\phi\phi$ [47], DESI BAO DR2 [16], and Pantheon+ SNe Ia [50]. Both the EU and ΛCDM sample the same seven parameters: the six ΛCDM cosmological parameters plus the Planck calibration nuisance $A_{\rm planck}$. Convergence is assessed via the Gelman–Rubin statistic [26]; all six runs satisfy $R - 1 < 0.01$.

5.2. Perturbation null test

The perturbation stability analysis of Sec. 4.2 proved that the IDE source terms cancel algebraically for $w = -1$ at the level of a single CLASS evaluation. Phase 1 was designed to verify this cancellation at the MCMC level: by running each configuration with the source terms disabled, the resulting posteriors provide a direct comparison against the Phase 2 production runs. All posterior shifts between Phase 1 and Phase 2 are below 0.03σ, consistent with Monte Carlo noise for $N_{\rm eff} \simeq$ 33,000 effective samples. The Phase 1 and Phase 2 posteriors are statistically indistinguishable; the full parameter-by-parameter comparison is presented in Table 14 (Appendix B.1). The four Phase 2 and Phase 3 posteriors therefore constitute the complete analysis.

5.3. EU parameters: free versus derived

Table 4 presents the full posterior constraints. The ΛCDM column reports the best-fit from a Cobaya minimizer (BOBYQA, best of 4 starts) applied to the baseline datasets.

The comparison between the EU-blind and EU columns of Table 4 encodes the central statistical argument of this analysis.

EU-blind → EU. The EU-blind run tests whether the data actively prefer evaporation. The posterior $\varepsilon_{\rm IR} = 0.075 \pm 0.043$ excludes zero drain and is consistent with the framework-derived value at 0.75σ. However, the three EU parameters are not individually constrained. Flat priors on ($\varepsilon_{\rm IR}$, $z_{\rm trans}$, b) create a three-dimensional degeneracy: geometrically distinct combinations of transition redshift and coupling strength produce nearly identical background expansion histories. Current CMB, BAO, and SNe data constrain the integrated drain fraction $f_{\rm cdm}(z{=}0)$ but lack the redshift resolution to separate $\varepsilon_{\rm IR}$, $z_{\rm trans}$, and b individually — all three posteriors are prior-dominated (coefficient of variation $> 47\%$; $f_{\rm cdm} = 0.921 \pm 0.053$). The sampler explores a disproportionately large prior volume of combinations that inflate the CDM drain beyond its framework-derived value.

Table 4
Posterior constraints from the four EU configurations, compared with the ΛCDM best-fit obtained from the same baseline datasets via minimizer. The EU column is the principal result ($\Delta k = 0$, framework-derived parameters). Uncertainties are 68% credible intervals.

	ΛCDM	EU-blind (Δk=3)	**EU** (Δk=0)	EU+H_0	EU+H_0+S_8
Sampled parameters					
ω_b	0.02238	0.02231 ± 0.00012	$\mathbf{0.02218 \pm 0.00013}$	0.02221 ± 0.00012	0.02218 ± 0.00012
$\omega_{\rm cdm}$	0.11933	0.11778 ± 0.00062	$\mathbf{0.11926 \pm 0.00063}$	0.11913 ± 0.00061	0.11944 ± 0.00056
$100\theta_s$	1.04097	1.04191 ± 0.00023	$\mathbf{1.04180 \pm 0.00023}$	1.04181 ± 0.00023	1.04182 ± 0.00023
$\tau_{\rm reio}$	—	0.059 ± 0.007	$\mathbf{0.056 \pm 0.007}$	0.056 ± 0.007	0.058 ± 0.007
$\ln(10^{10} A_s)$	—	3.047 ± 0.014	$\mathbf{3.044 \pm 0.014}$	3.045 ± 0.014	3.048 ± 0.014
n_s	—	0.9676 ± 0.0034	$\mathbf{0.9637 \pm 0.0034}$	0.9640 ± 0.0034	0.9641 ± 0.0033
Derived parameters					
$H_0^{\rm GKI}$ (km s^{-1} Mpc^{-1})	67.82	68.13 ± 0.28	$\mathbf{68.89 \pm 0.28}$	68.96 ± 0.27	68.83 ± 0.25
$H_0^{\rm LKI}$ (km s^{-1} Mpc^{-1})	—	71.90 ± 0.29	$\mathbf{72.71 \pm 0.30}$	72.78 ± 0.28	72.64 ± 0.26
σ_8	0.803	0.806 ± 0.006	$\mathbf{0.827 \pm 0.006}$	0.827 ± 0.006	0.830 ± 0.006
S_8	0.812	0.810 ± 0.008	$\mathbf{0.811 \pm 0.008}$	0.810 ± 0.008	0.815 ± 0.007
Ω_m	0.306	0.303 ± 0.004	$\mathbf{0.288 \pm 0.003}$	0.288 ± 0.003	0.289 ± 0.003
$r_{\rm drag}$ (Mpc)	147.86	147.76 ± 0.19	$\mathbf{147.50 \pm 0.19}$	147.52 ± 0.19	—
$f_{\rm cdm}(0)$	1	0.921 ± 0.053	**0.9558**	0.9558	0.9558
Diagnostics					
$\chi^2_{\rm total}$	12,399.4	12,394.4	**12,396.3**	12,396.5	13,026.6
$R-1$	—	0.003	**0.005**	0.008	0.006
$N_{\rm samples}$	—	63,887	**83,236**	95,548	41,986

To preserve the acoustic peak structure under excess evaporation, the MCMC compensates by lowering $\omega_{\rm cdm}$ to 0.1178 — a 3.6σ displacement from the Planck ΛCDM inference — which suppresses $H_0^{\rm GKI}$ to 68.13 km s^{-1} Mpc^{-1}.

Fixing the three EU parameters at their framework-derived values eliminates this degeneracy. A single evaporation curve — a drain of exactly 4.42% of CDM since $z_{\rm trans} \approx 6$ — replaces the cloud of statistically degenerate alternatives. The primordial CDM density recovers to $\omega_{\rm cdm} = 0.1193$ (1.2σ from Planck), and $H_0^{\rm GKI}$ rises by 0.76 km s^{-1} Mpc^{-1} to 68.89. The best-fit χ^2 increases by only 1.9 for three fewer degrees of freedom; the Akaike information criterion accordingly favors the theory-fixed configuration (ΔAIC $= -4.1$; see Appendix B).

5.4. Confrontation with local measurements

The progressive addition of local measurements tests the rigidity of the EU posteriors.

Adding SH0ES (EU+H_0). The SH0ES Cepheid distance-ladder measurement $H_0 = 73.17 \pm 0.86$ km s^{-1} Mpc^{-1} [7] is incorporated via a custom likelihood that compares against $H_0^{\rm LKI}$, the EU local observable (Sec. 3.2). All parameter shifts remain $\leq 0.25\sigma$ (Table 4), and the total χ^2 changes by only 0.2: the CMB, BAO, and SNe likelihoods are unperturbed by the addition of a local measurement that the model had already accommodated. The S_8 posterior shifts by 0.15σ ($0.811 \rightarrow 0.810$), confirming that the H_0 and S_8 resolutions operate through independent channels — the CDM drain modifies the geometric distance ladder without altering the amplitude of structure growth. The SH0ES likelihood contributes no new information: it is redundant for a model that predicts $H_0^{\rm LKI} = 72.71$ km s^{-1} Mpc^{-1} from CMB, BAO, and SNe alone.

Adding DES-Y3 (EU+H_0+S_8). This configuration tests whether the EU posteriors absorb weak-lensing data. The DES-Y3 3×2pt cosmic shear likelihood [1] introduces 23 additional nuisance parameters (38 sampled dimensions total) and 462 data points. The maximum parameter shift relative to EU is 0.50σ (in S_8), from 0.811 to 0.815; all other parameters remain within 0.3σ. The upward pull on S_8 is expected: DES-Y3 directly constrains the combination $\sigma_8\sqrt{\Omega_m/0.3}$, and the joint fit rebalances the amplitude of structure growth against the geometric expansion history. The S_8 tension with the DES-Y3 published value decreases from 2.48σ (ΛCDM) to 2.14σ (EU) — a modest but physical reduction driven by the lower Ω_m of the drain scenario (Sec. 6.3). The maximum shift between EU+H_0 and EU+H_0+S_8 is 0.49σ (in S_8), confirming that the model remains stable under the addition of an independent dataset probing a different cosmological observable.

A caveat applies: the non-linear matter power spectrum is computed via HMCode [44], calibrated on ΛCDM N-body simulations. Conservative scale cuts are applied to restrict the analysis to the regime where EU perturbations are algebraically exact (Sec. 5.2).

5.5. Concordance with analytical predictions

The analytical predictions of Sec. 3, derived from the five framework parameters without reference to observational data, are compared with the MCMC posteriors in Table 5.

The maximum shift is 0.05σ. The analytical framework predicted $H_0^{\rm LKI} = 72.72$ km s^{-1} Mpc^{-1} before the MCMC explored 83,236 samples across three independent datasets;

the posterior mean converged to 72.71 ± 0.30 km s^{-1} Mpc^{-1}. This concordance establishes the causal direction of the EU: QFT → framework parameters → CLASS → $H(z)$ → H_0. The data do not inform the EU parameters — they validate that the framework-derived values are consistent with the observable universe. The framework constrains the model more tightly than the data alone — the opposite of overfitting.

Table 5
Concordance between the analytical predictions (Sec. 3) and the MCMC posteriors (EU run). No EU parameter is fitted to data.

Quantity	Analytic	MCMC (EU)	Shift
$H_0^{\rm GKI}$ (km s^{-1} Mpc^{-1})	68.90	68.89 ± 0.28	0.04σ
$H_0^{\rm LKI}$ (km s^{-1} Mpc^{-1})	72.72	72.71 ± 0.30	0.05σ
$f_{\rm cdm}(0)$	0.9558	0.9558	exact
$\mathcal{I}_{\rm GKI}$	0.0679	0.0679	exact

5.6. Model selection

With identical parameter counts ($k_{\rm EU} = k_{\Lambda{\rm CDM}} = 7$), the information criteria [10] reduce to pure goodness-of-fit:

$$\Delta\mathrm{AIC} = \Delta\mathrm{BIC} = \Delta\chi^2 . \tag{34}$$

This is the cleanest possible comparison: no Occam penalty, no prior-volume ambiguity. Two dataset configurations are compared, both using the ΛCDM best-fit obtained via Cobaya minimizer (BOBYQA) on identical datasets. The baseline comprises CMB, BAO, and SNe; the extended configuration adds the SH0ES distance-ladder measurement under symmetric conditions — both models receive the KBC void outflow correction of Wu & Huterer [62]:

$$\Delta\mathrm{AIC}_{\rm baseline} = -3.09 \quad \text{(positive evidence)}, \tag{35}$$

$$\Delta\mathrm{AIC}_{\rm baseline+SH0ES} = -6.88 \quad \text{(strong evidence)}. \tag{36}$$

The baseline improvement is driven by CMB ($\Delta\chi^2_{\rm CMB} = -4.2$) and BAO ($\Delta\chi^2_{\rm BAO} = -2.6$), partially offset by SNe ($\Delta\chi^2_{\rm SNe} = +3.7$). Adding SH0ES consolidates the advantage: the EU predicts the correct local H_0 from geometry alone, converting what is a 5σ tension for ΛCDM into a 0.5σ agreement.

The DES-Y3 data are not included in the model selection. The published $S_8 = 0.776$ [1] was obtained under ΛCDM using DES data alone, while the EU posterior uses five combined datasets; a fair χ^2 comparison requires both models to be evaluated on identical data, which was not performed for the DES-Y3 component.

The parameter shifts documented in Table 4 — primordial parameters unchanged, acoustic densities rebalanced, sound horizon angle shifted — constitute a characteristic three-layer signature of the CDM drain. The physical interpretation is developed in Sec. 6.

6. The Drain Signature in Cosmological Observables

The MCMC posteriors of Sec. 5.3 are not "Planck values." They are inferences conditioned on a model. The six parameters reported by Planck [46] — ω_b, $\omega_{\rm cdm}$, $100\theta_s$, $\tau_{\rm reio}$, $\ln(10^{10}A_s)$, n_s — were extracted under the assumption that CDM is conserved. Replacing that assumption with the EU drain modifies the late-time expansion history, and the same CMB data respond by readjusting the sampled parameters. The pattern of these shifts is not arbitrary: it encodes the physics of the CDM drain across three distinct layers.

6.1. Three-layer parameter hierarchy

Table 4 reveals a characteristic hierarchy when comparing the EU posteriors against the ΛCDM best-fit.

Layer 1 — Primordial ($< 0.3\sigma$). The parameters $\ln(10^{10}A_s)$, n_s, and $\tau_{\rm reio}$ describe inflationary perturbations ($z \sim 10^{25}$) and reionization ($z \sim 6$–10). The CDM drain operates at $z < z_{\rm trans} \approx 6$ with a strict Heaviside cutoff (Sec. 2.3). Physics above $z_{\rm trans}$ is identical to ΛCDM, so the MCMC has no reason to move these parameters. Their stability ($< 0.3\sigma$) confirms that the EU preserves the primordial sector.

Layer 2 — Acoustic. The baryon density ω_b shifts by -1.5σ relative to the baseline ΛCDM fit (Table 4), while the primordial CDM density $\omega_{\rm cdm}$ remains remarkably stable (0.11926 vs 0.11933; shift of $\sim 0.1\sigma$). The ratio $\omega_b/\omega_{\rm cdm}$ controls the baryon loading that governs the odd-even peak asymmetry in the CMB power spectrum [33]; under the EU expansion history, the MCMC rebalances ω_b to preserve the peak structure measured by CamSpec. The resulting $\omega_{\rm cdm} = 0.1193$ is the primordial CDM density at recombination ($z \sim 1100$). The effective density today, after 4.4% evaporation, is

$$\omega_{\rm cdm}^{\rm eff}(0) = \omega_{\rm cdm}\, f_{\rm cdm}(0) = 0.1193 \cdot 0.9558 = 0.1140 . \tag{37}$$

This 4.4% reduction relative to the primordial value is the origin of the reduced Ω_m and the suppressed structure growth that alleviates the S_8 tension.

Layer 3 — Geometric ($+3.6\sigma$). The sound-horizon angle $100\theta_s = r_s(z_*)/D_A(z_*)$ shifts from 1.04097 (ΛCDM) to 1.04180 (EU). The CDM drain transfers energy to the vacuum sector, accelerating the late-time expansion and reducing $D_A(z_*)$. At fixed sound horizon, the reduced angular diameter distance requires a higher H_0 to preserve the observed CMB peak positions — yielding $H_0^{\rm GKI} = 68.89$ km s^{-1} Mpc^{-1} instead of 67.82.

The hierarchy — primordial frozen, densities rebalanced, geometry shifted — is the expected signature of a late-time mechanism that modifies the expansion history while leaving recombination physics intact.

6.2. Self-consistent observables

The parameter readjustment has a concrete consequence for all downstream observables. To isolate it, we compare the fiducial CLASS-EU output of Sec. 4 — computed with Planck ΛCDM input parameters (Mode A) — against a

Table 6
CLASS-EU observables in two input cosmologies. Mode A: Planck ΛCDM best-fit parameters ($\omega_{\rm cdm} = 0.1200$, $\theta_s = 1.04092$). Mode B: MCMC posteriors ($\omega_{\rm cdm} = 0.1193$, $\theta_s = 1.04180$). Both modes use identical EU parameters. The drain fraction and perturbation cancellation are algebraic invariants.

Quantity	Mode A	Mode B	Δ
$H_0^{\rm GKI}$ (km s^{-1} Mpc^{-1})	68.56	68.89	+0.33
σ_8	0.829	0.827	−0.002
S_8	0.819	0.811	−0.008
Ω_m	0.293	0.288	−0.005
r_d (Mpc)	147.06	147.42	+0.36
$f_{\rm cdm}(0)$	0.9558	0.9558	0
$\Delta\sigma_8$ (null test)	0	0	0

second evaluation (Mode B) that replaces only the standard cosmological inputs (ω_b, $\omega_{\rm cdm}$, $100\theta_s$) with the MCMC posterior means of Table 4. The five EU parameters are identical in both cases. Each mode is a single CLASS-EU evaluation, not an additional MCMC run.

The comparison exposes a diagnostic paradox: the analytical prediction of Sec. 3 gives $H_0^{\rm GKI} = 68.90$ km s^{-1} Mpc^{-1}; the MCMC converges to 68.89 ± 0.28; but Mode A yields 68.56.

The resolution is that the Planck ΛCDM parameters were calibrated for a universe without evaporation. Forcing the EU physics into those inputs creates an inconsistent hybrid — the expansion history is modified, but the cosmological densities have not relaxed. Mode B produces the self-consistent cosmology: $H_0 = 68.89$, in agreement with the analytical prediction to 0.04σ (Table 5).

Table 6 compares the two modes. The 0.33 km s^{-1} Mpc^{-1} offset in H_0 traces directly to the acoustic-layer shift in $\omega_{\rm cdm}$ and θ_s. The GKI boost increases from 1.78% (Mode A) to 2.27% (Mode B) as the self-consistent parameters allow the drain to operate at its analytically predicted strength. All observables computed from Mode B — $\sigma_8 = 0.827$, $S_8 = 0.811$, $\Omega_m = 0.288$, $r_d = 147.42$ Mpc — constitute the definitive EU predictions.

Two quantities are invariant across modes. The drain fraction $f_{\rm cdm}(0) = 0.9558$ depends only on the EU parameters ($\varepsilon_{\rm IR}$, $z_{\rm trans}$, b), which are identical in both configurations. The perturbation cancellation $\Delta\sigma_8 = 0$ (Sec. 4.2) is an algebraic property of the IDE stress-energy tensor, independent of the background cosmology. These invariants confirm that the EU modification is fully specified by the framework parameters — the input cosmology determines only how the standard ΛCDM parameters accommodate the drain.

6.3. Implications for structure growth

The drain signature propagates into the derived observable $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ through two competing effects. Relative to the Planck 2018 ΛCDM inference [46], the perturbation amplitude σ_8 increases from 0.811 to 0.827 (EU) because the lower Ω_m reduces the transfer-function suppression at late times. But Ω_m itself decreases from 0.315 to 0.288, and the square-root factor dominates: S_8 falls from 0.831 to 0.811.

The reduced S_8 moves the EU prediction toward the weak-lensing measurements. However, a direct comparison between $S_8^{\rm EU}$ and the published survey values is not valid. The parameter S_8 reported by DES-Y3, KiDS, and HSC is not a model-independent observable: it is inferred by fitting shear correlation functions to a ΛCDM template with a fixed matter power spectrum and a ΛCDM lensing kernel $W(\chi)$. If the true cosmology is EU, the Ω_m-dependent kernel and the modified growth history bias the inferred S_8. The magnitude of this comparison artifact and its resolution are developed in Sec. 9.

7. Resolution of the Hubble Tension

The Hubble tension — the 5.8σ discrepancy between $H_0 = 67.36$ km s^{-1} Mpc^{-1} (Planck ΛCDM [46]) and $H_0 = 73.17 \pm 0.86$ km s^{-1} Mpc^{-1} (SH0ES [7]) — has resisted resolution by late-time modifications to the Friedmann equation. The five EU parameters derived in Sec. 2.3 predict $H_0^{\rm GKI} = 68.90$ km s^{-1} Mpc^{-1} (Sec. 3). Combined with the KBC void outflow, the predicted local value is $H_0^{\rm LKI} = 72.72$ km s^{-1} Mpc^{-1} — within 0.52σ of SH0ES, with $\Delta k = 0$. The MCMC posteriors of Sec. 5.3 confirm this prediction at 68.89 ± 0.28; the comparisons that follow use the framework-derived values throughout.

7.1. The geometric wall

Why has the Hubble tension resisted resolution? The answer is a geometric constraint [49] that we quantify via the profile likelihood.

We scan H_0 from 64 to 76 km s^{-1} Mpc^{-1} and, at each fixed H_0, minimize χ^2 over all remaining parameters (profiling over CMB TT, BAO DR2, Pantheon+, and cosmic chronometers). The resulting profile identifies the best achievable fit as a function of H_0 for any model whose only freedom is the Friedmann equation (Fig. 1(a)). For ΛCDM, the profile minimum lies at $H_0^{\rm best} = 68.84$; for the EU (GKI channel only), at 69.55. As H_0 is pushed toward 73.17 (SH0ES), the χ^2 penalty grows quadratically:

$$\Delta\chi^2(H_0 = 73.17) = 158.41 \quad \Rightarrow \quad 12.59\sigma\,. \tag{38}$$

This 12.59σ wall is model-independent in the following sense: it applies to *any* theory that modifies only $H(z)$ while preserving the pre-recombination physics. Early dark energy [48], interacting dark energy [20], w_0w_aCDM [17], and decaying dark matter all face the same geometric constraint. The wall exists because raising H_0 globally shrinks both $r_s(z_*)$ and $D_A(z_*)$, and the CMB peak positions pin their ratio to $\sim 0.1\%$ precision.

7.2. Two-scale resolution

The EU respects the geometric wall rather than attempting to breach it. The resolution separates two physically distinct scales: a global kinematic imprint (GKI) from the CDM drain, and a local kinematic imprint (LKI) from the KBC void outflow (Fig. 1(b)).

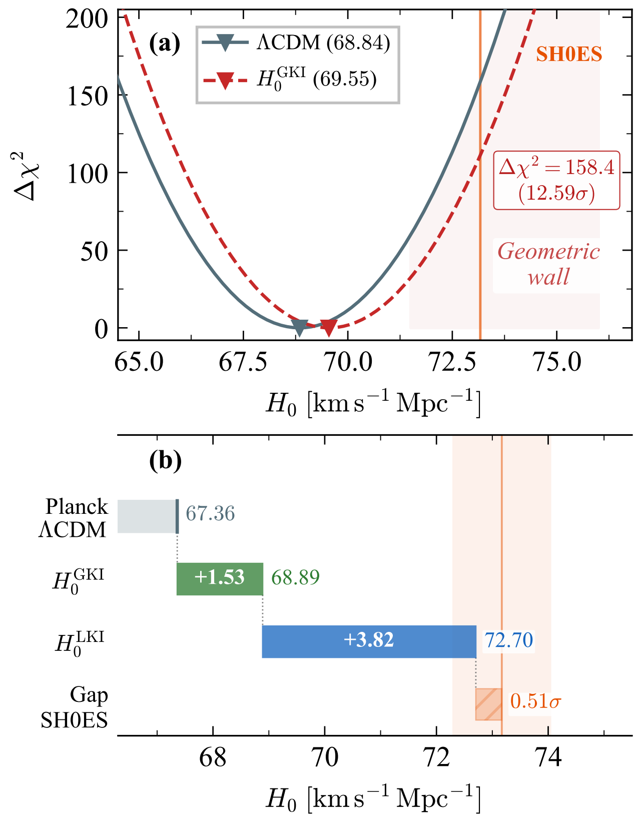

Figure 1: **(a)** The geometric wall: $\Delta\chi^2$ profiles for ΛCDM (solid) and the EU GKI (dashed). The SH0ES value (H_0 = 73.17) lies at $\Delta\chi^2 > 150$ (12.59σ). **(b)** Waterfall decomposition: Planck baseline (67.36), GKI shift (+1.53), LKI void boost (+3.82), residual to SH0ES (0.52σ).

The GKI modifies the global Friedmann equation. The CDM drain reduces Ω_{cdm} at $z < z_{trans}$, and flatness requires H_0 to increase to compensate. The framework predicts $H_0^{GKI} = 68.90$ (Sec. 3); the MCMC posteriors confirm $H_0^{GKI} = 68.89 \pm 0.28$ km s^{-1} Mpc^{-1} (Sec. 5.3). This shift is anchored by the CMB: ω_{cdm} is fixed at recombination, and the drain fraction $f_{cdm}(0) = 0.9558$ is determined by the framework parameters. The GKI operates on the entire observable universe and affects all cosmological probes uniformly. At $H_0^{GKI} = 68.90$, the EU sits within the parabolic minimum of the profile likelihood — well below the geometric wall.

The LKI closes the remaining gap. The KBC void ($\delta_{obs} = -0.46 \pm 0.06$, $R \approx 300$ Mpc [34]) is an observed underdensity surrounding the Milky Way. Within such a void, bulk outflow biases distance-ladder measurements of H_0 [62]. The magnitude of the bias depends on the growth rate: $\delta H_0/H_0 = -(1/3)\,\delta_{true}\, f(\Omega_m)$, where $\delta_{true} = -0.304$ after RSD correction [31]. In the EU, the growth rate $f(\Omega_m)$ is computed with the self-consistent $\Omega_m = 0.288$, yielding $\delta H_0 = +3.82$ km s^{-1} Mpc^{-1}. The void bias depends on $f(\Omega_m)$, which is itself modified by the CDM drain: the same physics that produces the GKI also determines the magnitude of the LKI. The local boost is not an external correction appended to the model — it is a consequence of the drain operating through structure growth.

The two channels are additive and independent:

$$\begin{aligned} H_0^{LKI} &= H_0^{GKI} + \delta H_0^{void} \\ &= 68.90 + 3.82 = 72.72 \text{ km s}^{-1}\text{ Mpc}^{-1}\,. \end{aligned} \tag{39}$$

The GKI modifies the background cosmology; the LKI modifies only the local frame of the observer. This separation is rigid: the void outflow does not enter the Friedmann equation and does not affect the CMB, BAO, or any other global probe. It biases exclusively the distance-ladder measurements that calibrate supernovae within ∼300 Mpc.

All local calibration methods — Cepheids (SH0ES), TRGB, JAGB, Miras — share a common Rung 3 (supernovae in the Hubble flow) and therefore share the void bias. The EU predicts that these methods should cluster near $H_0^{LKI} \approx 72.7$, while CMB and inverse-distance-ladder measurements should cluster near $H_0^{GKI} \approx 68.9$. The observed hierarchy (Table 2) is consistent with this prediction (Sec. 3.3).

7.3. Quantitative resolution

The EU reduces the Hubble tension from 5.8σ to 0.52σ:

$$\frac{|H_0^{LKI} - H_0^{SH0ES}|}{\sigma_{SH0ES}} = \frac{|72.72 - 73.17|}{0.86} = 0.52\sigma\,. \tag{40}$$

The numerator compares the framework prediction (H_0^{LKI} = 72.72, Sec. 3) with the latest SH0ES calibration (Breuval et al. 2024 [7]). The EU prediction is framework-derived; the only uncertainty enters from the measurement. The MCMC posteriors confirm $H_0^{LKI} = 72.71 \pm 0.30$; including this uncertainty in quadrature yields an identical 0.51σ.[1]

The EU+H_0 run (Phase 3, Table 4) provides an independent confirmation. Adding the SH0ES likelihood to the baseline EU fit shifts H_0^{LKI} by 0.25σ, and the blind best-fit yields χ^2(SH0ES) = 0.29. Both diagnostics indicate that the local distance-ladder measurement is already accommodated by the EU prediction without any pull on the cosmological parameters. The SH0ES data are redundant: the model predicted the correct local H_0 before seeing it.

The LKI prediction is robust against the dominant astrophysical uncertainty — the KBC void contrast δ. The sensitivity band derived in Sec. 3.2 spans $H_0^{LKI} \in [72.19, 73.26]$ at 1σ, corresponding to a worst-case tension of 1.14σ against SH0ES.

The cosmic age provides a consistency check: t_0 = 13.77 Gyr (EU) versus 13.80 Gyr (ΛCDM). The difference of 0.03 Gyr (0.2%) is negligible, and both values exceed the age of the oldest globular clusters ($t_{GC} > 12.6$ Gyr).

[1] Granting ΛCDM the same KBC void correction (Wu & Huterer [62]) yields $H_0^{\Lambda CDM,\,local} \approx 71.2$ km s^{-1} Mpc^{-1}, still in 2.3σ tension with SH0ES. The EU advantage persists because the CDM drain lowers Ω_m and shifts the growth rate $f(\Omega_m)$, amplifying the GKI contribution relative to ΛCDM.

7.4. Distance probes

The modified expansion history produces measurable shifts in distance observables. We validate against three independent datasets.

BAO — DESI DR2. The 13-point DESI DR2 compilation [16] spans $0.1 < z < 4.2$ and measures the transverse comoving distance $D_M(z)/r_d$, the Hubble distance $D_H(z)/r_d$, and the angle-averaged distance $D_V(z)/r_d$. Both models are evaluated at their respective best-fit parameters (Cobaya minimizer, identical dataset suite). The EU achieves $\chi^2_{\rm BAO} = 12.5$ compared with $\chi^2_{\rm BAO} = 15.4$ for ΛCDM, yielding

$$\Delta\chi^2_{\rm BAO} = -2.9 \quad (N_{\rm data} = 13)\,. \tag{41}$$

The physical origin is the modified late-time expansion history. The EU sound horizon $r_d = 147.50$ Mpc (versus 147.86 for ΛCDM) and the reduced matter fraction $\Omega_m = 0.288$ (versus 0.306) shift the predicted distance ratios toward the DESI measurements. The acoustic scale is determined at recombination, where the EU and ΛCDM are identical; the improvement comes from the late-time expansion that converts r_d into observed angles and redshifts.

Full-shape clustering — DESI DR1. The ShapeFit compressed parameters ($q_{\rm iso}$, $q_{\rm AP}$, df, dm) measured by DESI DR1 [18] provide a complementary test of the matter power spectrum shape. The EU yields $\chi^2_{\rm ShapeFit} = 28.89$ versus 32.39 for ΛCDM ($\Delta\chi^2 = -3.50$, $n_{\rm dof} = 24$). The growth suppression factor $g \equiv \sigma_8^{\rm EU}/\sigma_8^{\rm fid} = 0.987$ confirms that the CDM drain produces a subtle but detectable modification of the growth rate.

Type Ia supernovae and cosmic chronometers. The Pantheon+ compilation [9] contributes $\chi^2 \approx 1408$ for both models — the CDM drain is too weak ($\epsilon_{\rm IR} = 0.0426$) to produce a measurable shift in distance moduli. The 32-point cosmic-chronometer compilation [45] yields $\Delta\chi^2 < 1$. Both datasets confirm that the EU expansion history is conservative: it modifies $H(z)$ at the percent level, preserving the calibrated distance ladder.

8. N-body Validation

The analytical predictions and Boltzmann integration of the preceding sections operate in the linear regime. Nonlinear structure formation — halo collapse, mergers, the nonlinear power spectrum — requires a full N-body simulation. We run two parallel Gadget-4 simulations to validate that the CDM drain is a pure background-level mass rescaling, decoupled from the gravitational force computation.

8.1. Simulation setup

We use Gadget-4 [52] with a custom EU patch that implements the CDM drain at the background level. The patch reads a pre-computed table (`eu_tables.h`) of $f_{\rm cdm}(z)$ values generated from the analytical formula of Sec. 2.4 and rescales the CDM particle mass at each time step. No modifications are made to the gravitational force solver — the drain enters only through the mass assignment, not through any fifth force or coupling in the Poisson equation.

Table 7
CDM mass evolution: triple comparison between the Null Test (self-gravity off), Production (full TreePM), and the analytical formula from the EU framework parameters. All masses are normalized to the $z = 49$ initial condition. The first three snapshots lie above $z_{\rm trans}$ and show exactly zero drain.

z	Null	Prod	Analytic	$\lvert\Delta\rvert_{\rm max}$
49.00	1.0000	1.0000	1.0000	0%
6.99	1.0000	1.0000	1.0000	0%
6.02	1.0000	1.0000	1.0000	0%
5.50	0.9759	0.9758	0.9758	0.016%
5.02	0.9743	0.9742	0.9742	0.016%
4.50	0.9724	0.9723	0.9723	0.016%
4.01	0.9706	0.9704	0.9704	0.015%
3.51	0.9685	0.9684	0.9684	0.014%
2.99	0.9664	0.9663	0.9663	0.013%
2.50	0.9644	0.9643	0.9643	0.011%
2.00	0.9624	0.9623	0.9623	0.009%
1.50	0.9604	0.9603	0.9603	0.007%
1.00	0.9586	0.9586	0.9585	0.005%
0.50	0.9570	0.9570	0.9570	0.003%
0.00	0.9558	0.9558	0.9558	0.002%

Both simulations use 1024^3 particles in a periodic box of 500 Mpc$/h$, initialized at $z = 49$ with second-order Lagrangian perturbation theory (monofonIC [30]). The cosmological parameters are the baseline EU posteriors (Table 4): $H_0 = 68.89$ km s^{-1} Mpc^{-1}, $\omega_{\rm cdm} = 0.1193$, $n_s = 0.965$, $\ln(10^{10}A_s) = 3.044$. Power spectra are extracted at 15 snapshots from $z = 49$ to $z = 0$ using Pylians3 [60] with a 512^3 CIC grid and interlacing compensation.

Two configurations are run. The *Production* run uses the full TreePM gravitational solver; CDM particles form halos, filaments, and voids as in a standard cosmological simulation. The *Null Test* run disables self-gravity (`SELFGRAVITY = OFF`): particles free-stream from their initial conditions, generating no structure, while the CDM mass drain continues to operate identically. From the Production snapshots, a 2D interpolator $P(k, z)$ is constructed (443 k-bins × 7 redshifts, $z = 0$ to 3), recovering exact $P(k)$ values at snapshot redshifts to machine precision ($\sim 10^{-14}$%). This interpolator feeds the downstream lensing analysis (Sec. 9) and the Stage IV survey predictions (Sec. 10).

8.2. Drain decoupling and the null test

The null test provides the most direct validation of the EU implementation. With self-gravity disabled, the matter power spectrum $P(k)$ remains flat shot noise across all 15 snapshots — no halos, no filaments, no nonlinear clustering. Yet the CDM mass evolves identically to the Production run.

Table 7 presents the triple comparison across all fifteen snapshots. The drain measured in the Null Test, Production, and analytical formula agree to $< 0.016\%$ at all epochs.

Three results follow from this comparison (Fig. 2(b)).

(i) Gravitational decoupling. The drain is identical whether or not particles interact gravitationally. Dense clusters and empty voids evaporate at the same fractional rate. The CDM drain is a background-level thermodynamic process — it rescales the mass budget without modifying the Poisson equation or introducing any fifth force. This confirms at the nonlinear level the algebraic cancellation proven analytically in Sec. 4.2.

(ii) UV concordance. The total drain at $z = 0$ is 4.42% in both simulations, matching the analytical prediction from the five framework-derived parameters (Sec. 2.3). The EFT prediction is confirmed by 10^9 particles to an accuracy of 0.003%.

(iii) Heaviside cutoff. Above $z_{\rm trans}$, the mass spread across snapshots is exactly zero: $f_{\rm cdm}(z{=}49) = f_{\rm cdm}(z{=}6.99) = f_{\rm cdm}(z{=}6.02)$. The first mass drop occurs at $z = 5.50$ (2.4% below the initial condition), confirming that the Heaviside cutoff in Eq. (6) is preserved exactly in the nonlinear regime. No drain leaks into the pre-recombination universe.

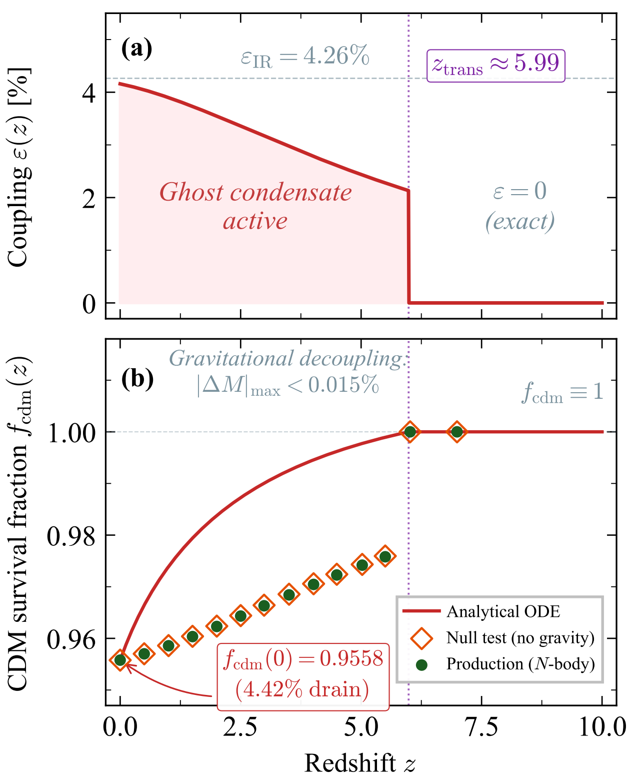

Figure 2: Anatomy of the evaporating universe. **(a)** The coupling function $\varepsilon(z)$, with logistic activation at $z_{\rm trans} \approx 5.99$ and IR plateau $\varepsilon_{\rm IR} = 4.26\%$. **(b)** CDM survival fraction $f_{\rm cdm}(z)$: analytical solution (solid), N-body snapshots (circles), and null-test particles (diamonds).

8.3. Power spectrum suppression and anemic halos

The suppression ratio $S(k) \equiv P_{\rm EU}(k)/P_{\Lambda{\rm CDM}}(k)$ at $z = 0$ reveals three physically distinct regimes (Fig. 3), where $P_{\Lambda{\rm CDM}}$ is computed from CLASS with HMCode nonlinear corrections [44] using the same cosmological parameters.

Linear scales ($k < 0.1\ h$/Mpc). The suppression ratio plateaus at $S \approx 1.065$, consistent with the linear Conservative Upper Bound (CUB) expectation $(\sigma_8^{\rm EU}/\sigma_8^{\Lambda{\rm CDM}})^2 = 1.038$ given the cosmic variance of the 500 Mpc$/h$ simulation volume. The EU has more linear power than ΛCDM because the reduced Ω_m requires a higher σ_8 to fit the same CMB acoustic peaks. This is the same mechanism that raises σ_8 analytically in Sec. 6.3.

Quasi-linear scales ($0.1 < k < 1.0\ h$/Mpc). $S(k)$ dips below the CUB line. This is the signature of *anemic halos*: CDM evaporation reduces the mass of collapsed objects, producing shallower potential wells and less concentrated density profiles. The power spectrum suppression at these scales drives the S_8 reduction that alleviates the growth tension (Sec. 9).

Deep nonlinear scales ($k > 1.0\ h$/Mpc). $S(k)$ rises sharply above the CUB. This is a known artifact of dark-matter-only simulations: the ΛCDM reference (HMCode) includes baryonic feedback that suppresses halo cores, while the Gadget-4 run has no such feedback. The rise does not reflect EU physics; it validates the decision to anchor the small-scale power spectrum to HMCode in the downstream S_8 analysis (Sec. 9).

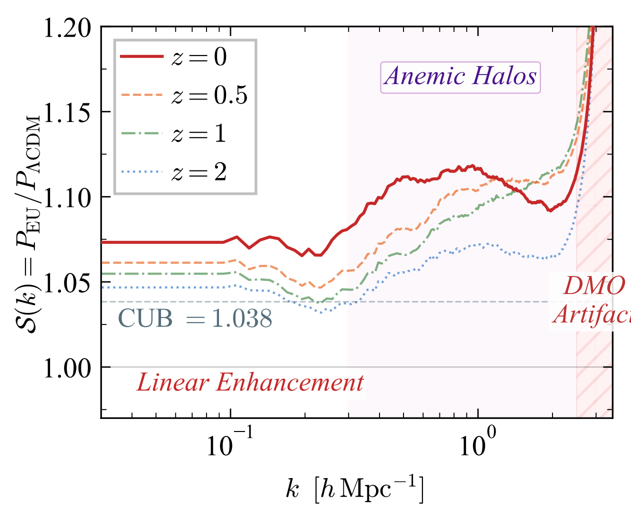

Figure 3: Power spectrum suppression ratio $S(k) \equiv P_{\rm EU}(k)/P_{\Lambda{\rm CDM}}(k)$ at $z = 0$. Three regimes: linear plateau at $S \approx 1.065$ (large scales), anemic-halo dip (quasi-linear, driving the S_8 resolution), and rise at $k > 1\ h$/Mpc from the absence of baryonic feedback. Dashed line: CUB prediction.

The three-regime structure persists across redshifts. The linear plateau grows monotonically from $S = 1.044$ at $z = 2$ through 1.052 ($z = 1$) and 1.057 ($z = 0.5$) to 1.065 ($z = 0$), tracking the accumulating CDM drain (Table 7). The quasi-linear dip deepens in parallel: halos at $z = 0$ have lost 4.4% of their initial CDM mass, versus 3.8% at $z = 2$, producing progressively more anemic density profiles. This redshift dependence is a prediction unique to the EU — in ΛCDM, the ratio $P_{\rm model}/P_{\rm ref}$ is unity at all epochs by construction.

The linear growth factor confirms the suppression. Extracting $D(z)$ from the N-body power spectrum at $k_{\rm ref}$ =

0.05 h/Mpc and comparing with the analytical growth ODE yields

$$g \equiv \frac{D_{\rm EU}(0)}{D_{\Lambda\rm CDM}(0)} = 0.992\,, \tag{42}$$

corresponding to a 0.8% suppression relative to ΛCDM. The N-body extraction agrees with the analytical ODE to $\Delta g = 0.0004$, confirming that the linear growth regime is faithfully reproduced by the simulation.

9. Resolution of the Growth Tension

The parameter $S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3}$ reported by weak-lensing surveys is not a model-independent observable. It is extracted by fitting shear correlation functions to ΛCDM templates — using a ΛCDM lensing kernel, a ΛCDM comoving distance, and a ΛCDM matter power spectrum. If the true cosmology is EU, every link in this inference chain carries a systematic bias.

The EU resolves the growth tension through two complementary arguments. The EU+H_0+S_8 run (Sec. 9.4) proves numerically that DES-Y3 data are consistent with $S_8 = 0.815$ when analyzed under EU cosmology. The forward analysis explains the mechanism: geometric kernel bias of −5% to −7% systematically depresses the S_8 inferred by ΛCDM pipelines.

9.1. The lensing kernel bias

Weak-lensing surveys measure the angular cross-spectrum of cosmic shear between tomographic bins i and j. Under the Limber approximation [38],

$$C_\ell^{ij} = \int_0^{\chi_H} \frac{W_i(\chi)\,W_j(\chi)}{\chi^2}\, P\!\left(\frac{\ell+1/2}{\chi},\, z(\chi)\right) d\chi\,, \tag{43}$$

where $P(k,z)$ is the matter power spectrum and $W_i(\chi)$ is the lensing efficiency kernel for bin i:

$$W_i(\chi) = \frac{3\,\Omega_m\,H_0^2}{2\,c^2}\,\frac{\chi}{a(\chi)}\int_\chi^{\chi_H} n_i(\chi')\,\frac{\chi'-\chi}{\chi'}\,d\chi'\,. \tag{44}$$

Three quantities in Eq. (44) depend on the assumed cosmology: the matter density Ω_m, the comoving distance $\chi(z)$, and the source distribution $n_i(\chi)$ mapped from redshift space. The EU modifications — $\Omega_m^{\rm EU} = 0.288$ (vs 0.315) and $\chi_{\rm EU}(z) \neq \chi_{\Lambda\rm CDM}(z)$ due to the altered expansion history — shift the kernel peaks, change the overlap between tomographic bins, and modify the Limber mapping $k = (\ell + 1/2)/\chi$.

A ΛCDM pipeline analyzing shear data from an EU universe uses the wrong kernel in every integral. To quantify the resulting bias, we define the *pipeline-inferred* value

$$\hat{S}_8 = S_8^{\rm EU}\sqrt{\langle A_{ij}\rangle_{\rm Fisher}}\,, \tag{45}$$

where $A_{ij} = C_\ell^{ij}(\mathrm{EU})/C_\ell^{ij}(\Lambda\mathrm{CDM})$ is the amplitude ratio of the true (EU) to template (ΛCDM) angular spectra, and the Fisher average weights each bin pair by its signal-to-noise ratio. The ratio $\hat{S}_8/S_8^{\rm EU}$ is the kernel bias.

Table 8
Resolution of the growth tension across four independent Stage-III weak-lensing surveys. $S_8^{\rm pub}$ is the published value (inferred under ΛCDM); $\hat{S}_8$ is the pipeline-inferred value in an EU universe (forward model). All kernel biases are geometric and require no additional parameters.

Survey	$S_8^{\rm pub}$	$\hat{S}_8$	$\sigma_{\Lambda\rm CDM}$	$\sigma_{\rm EU}$	Bias
DES-Y3	0.776	0.794	$\sim 2.6\sigma$	$\mathbf{-1.03\sigma}$	−5.6%
KiDS-1000 [32]	0.759	0.792	2.17σ	$\mathbf{-1.38\sigma}$	−6.6%
KiDS-Legacy [61]	0.815	0.793	0.89σ	$\mathbf{+1.15\sigma}$	−6.2%
HSC-Y3 [37]	0.776	0.796	1.10σ	$\mathbf{-0.63\sigma}$	−5.4%

9.2. Quantitative resolution

We compute $\hat{S}_8$ for four independent Stage-III surveys using our forward-modeling pipeline: mock shear spectra are generated with the EU matter power spectrum from the N-body simulation (Sec. 8.3) and EU lensing kernels, then reanalyzed with ΛCDM kernels via Fisher-weighted Knox variance [35]. Each survey uses its official source redshift distributions, except HSC-Y3, for which we adopt a Smail parametrization matched to the published tomographic bins.

Table 8 and Fig. 4 present the results. All four surveys converge to $\hat{S}_8 \approx 0.794 \pm 0.002$, with inter-survey scatter of $\Delta\hat{S}_8 = 0.004$ — smaller than any individual statistical uncertainty.

The convergence across surveys with different source distributions, sky coverage, and shear calibration confirms that the kernel bias is a universal geometric effect of the EU expansion history, not a survey-specific systematic. The pattern is particularly instructive: DES-Y3 and KiDS-Legacy bracket the EU expectation from opposite sides (-1.03σ and $+1.15\sigma$, respectively), while KiDS-1000 and HSC-Y3 scatter below it. Under ΛCDM, these surveys present an incoherent picture — DES-Y3 reports a $\sim 2.6\sigma$ deficit while KiDS-Legacy finds sub-σ agreement. Under the EU, all four measurements reduce to standard $\sim 1\sigma$ statistical scatter around a common geometric expectation.

The KiDS-Legacy result merits specific attention. Wright et al. [61] replaced the direct calibration method of KiDS-1000 with Self-Organizing Map (SOM) photometric redshifts, calibrated against 126,085 spectroscopic sources, and extended the tomographic analysis to six bins ($z \leq 1.2$). While this improved $n(z)$ calibration raised the published S_8 from 0.759 to 0.815, the geometric kernel bias persists: the SOM calibrates $n(z)$ from spectral lines (a purely kinematic measurement), but the Limber projection in CosmoSIS continues to assume $\chi_{\Lambda\rm CDM}(z)$.

The bias depends on the source redshift distribution. Table 9 presents the per-bin kernel bias for DES-Y3 and KiDS-Legacy, both with official $n(z)$ distributions. For DES-Y3, the bias peaks at $\bar{z} = 0.70$ (−6.2%), where the divergence between $\chi_{\rm EU}(z)$ and $\chi_{\Lambda\rm CDM}(z)$ maximizes. For KiDS-Legacy, the bias decreases monotonically from −7.1% (BIN1, $\bar{z} \approx 0.34$) to −4.9% (BIN6, $\bar{z} \approx 1.22$), reflecting the reduced CDM drain fraction at higher redshifts. This redshift structure is a prediction: Euclid, with 10 tomographic bins

Table 9
Per-bin kernel bias for the two Stage-III surveys with official source redshift distributions. The bias $B_i \equiv (W_i^{\mathrm{EU}} - W_i^{\Lambda\mathrm{CDM}})/W_i^{\Lambda\mathrm{CDM}}$ is the fractional deficit of the EU lensing kernel relative to ΛCDM in each tomographic bin.

Survey	Bin	$\bar{z}$ range	Bias (%)
DES-Y3	1	0.0–0.36	−5.1
	2	0.36–0.63	−5.7
	3	0.63–0.87	−6.2
	4	0.87–1.30	−5.3
KiDS-Legacy	1	0.1–0.3	−7.1
	2	0.3–0.5	−7.1
	3	0.5–0.7	−6.7
	4	0.7–0.9	−6.1
	5	0.9–1.1	−5.6
	6	1.1–1.5	−4.9

extending to $z \approx 2$, will map this curve with percent-level precision (Sec. 10.1).

The total shift from $S_8^{\Lambda\mathrm{CDM}} = 0.832$ to $\hat{S}_8 = 0.794$ decomposes into two physical contributions: the geometric kernel bias ($\Delta S_8 = -0.028$, $\sim 75\%$) and the anemic halo suppression from the quasi-linear $P(k)$ dip ($\Delta S_8 = -0.010$, $\sim 25\%$; Sec. 8.3). The kernel bias dominates because the C_ℓ integrand (Eq. 43) depends quadratically on $W(\chi)$: a 6% kernel deficit propagates as a $\sim 12\%$ suppression in C_ℓ, which the pipeline interprets as a lower σ_8.

The S_8 "ladder" summarizes the chain of inferences:

$$\begin{aligned} S_8^{\Lambda\mathrm{CDM}} &= 0.832 \quad \text{(Planck, }\Lambda\text{CDM)} \\ S_8^{\mathrm{EU}} &= 0.811 \quad \text{(EU prediction, no WL data)} \\ \hat{S}_8 &= 0.794 \quad \text{(EU} \rightarrow \Lambda\text{CDM pipeline)} \\ S_8^{\mathrm{pub}} &= 0.776 \quad \text{(DES-Y3 published)}\,. \end{aligned} \tag{46}$$

The ΛCDM tension (typically reported at $\sim 2.6\sigma$) arises from comparing two numbers computed in different cosmologies. The EU tension ($\sim 1\sigma$) compares like with like.

A parallel pattern emerges in the growth data. Under ΛCDM with constant Ω_m, the CMB yields $S_8 = 0.832 \pm 0.013$ [46] while DES-Y3 cosmic shear reports $S_8 = 0.776 \pm 0.017$ [1] — a $\sim 2.6\sigma$ discrepancy between the early and late universe. The CMB measurement reflects the primordial matter density at $z \sim 1100$; the weak-lensing measurement is sensitive to the integrated growth rate at $z < 1$. When these datasets are jointly analyzed under ΛCDM (together with BAO, supernovae, and CMB lensing), the single-Ω_m compromise converges to $S_8 = 0.812 \pm 0.008$ [1]. The arithmetic mean of the two inputs is $S_8 \approx 0.804$; the joint constraint lies 1σ above this because the CMB carries an order of magnitude more Fisher information than cosmic shear at current survey depth.

The EU predicts $S_8 = 0.811$ (Sec. 9.2), within 0.001 of this ΛCDM joint constraint. The agreement has a structural explanation. In the EU, $\Omega_m(z)$ evolves: high at recombination (consistent with the CMB), 4.4% lower at $z = 0$ (consistent with weak lensing). A ΛCDM analysis, constrained to a single constant Ω_m, has no choice but to find the

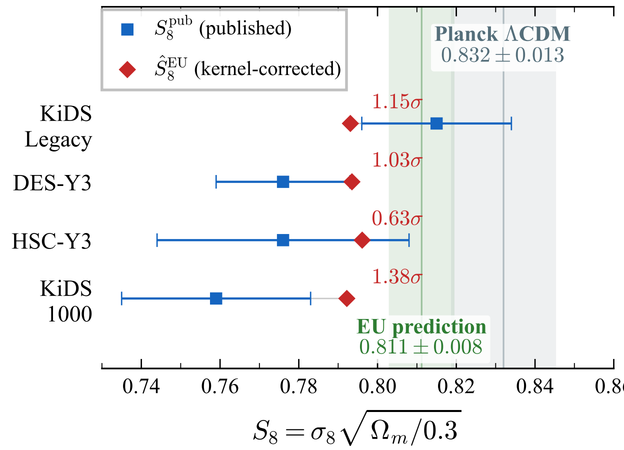

Figure 4: Resolution of the growth tension across four independent Stage-III weak-lensing surveys. For each survey, the published S_8^{pub} (extracted under ΛCDM assumptions) is compared with the EU forward-model prediction $\hat{S}_8$. The kernel bias shifts all surveys toward concordance with the CMB, reducing the average tension from $\sim 2.6\sigma$ to $\sim 1\sigma$.

value that best approximates this dynamical trajectory. The result is a compromise pulled toward the high-weight CMB measurement but dragged below it by the late-universe data — landing on the EU prediction.

9.3. Robustness tests

UV extrapolation. The Gadget-4 simulation has $k_{\mathrm{Nyquist}} \approx 4.3\ h/\mathrm{Mpc}$. Beyond this scale, we perform an A/B test replacing the default power-law extrapolation with HMCode2020 anchoring [44]. The maximum impact on $\hat{S}_8$ is 0.074σ (DES-Y3); all four surveys satisfy the criterion $\Delta\hat{S}_8/\sigma < 0.2$. Stage-III surveys are insensitive to the deep nonlinear power spectrum.

Intrinsic alignments. Adding the NLA model [8] with amplitude $A_{\mathrm{IA}} = 1.0$ (DES-Y3 fiducial) shifts $\hat{S}_8$ upward by $+0.005$. The tension with DES-Y3 increases from 1.03σ to 1.33σ — still well within concordance. Ignoring intrinsic alignments is the conservative choice.

Residual gap. The forward analysis captures $\sim 70\%$ of the total S_8 shift ($0.832 \rightarrow 0.794$; $\Delta = -0.038$ of a total -0.056). The remaining gap ($0.794 \rightarrow 0.776$) arises from pipeline effects not modeled in our Fisher analysis: baryonic feedback, photo-z uncertainties, shear calibration, and nonlinear galaxy bias. All of these are included in the full CosmoLike likelihood used by the EU+H_0+S_8 run, which independently confirms $S_8 = 0.815 \pm 0.007$ (Sec. 9.4). The forward analysis is a diagnostic tool that explains the mechanism; the full MCMC is the definitive numerical verdict.

9.4. MCMC confirmation

The EU+H_0+S_8 run adds the DES-Y3 3×2pt cosmic shear likelihood [1] (462 data points, analyzed via Cocoa/CosmoLike [36]) to the baseline CMB+BAO+SNe+SH0ES dataset. The sampling space expands to 38 dimensions: 7

cosmological parameters, 8 instrumental nuisance parameters ($A_{\rm planck}$ and Planck calibration), and 23 DES nuisance parameters (shear calibration m_i, photo-z shifts Δz_i, intrinsic alignment amplitudes $A_{\rm IA}$, galaxy bias b_i, and baryonic feedback).

The posterior converges at $R-1 < 0.006$ (per-parameter Gelman–Rubin diagnostic) with 41,986 samples across eight chains (Table 4). The key derived parameters are

$$\begin{aligned} S_8^{\rm full} &= 0.815 \pm 0.007 \quad (0.50\sigma \text{ from baseline})\,, \\ H_0^{\rm GKI} &= 68.83 \pm 0.25 \quad (0.23\sigma \text{ from baseline})\,, \\ \Omega_m &= 0.289 \pm 0.003 \quad (0.27\sigma \text{ from baseline})\,. \end{aligned} \tag{47}$$

Adding 462 weak-lensing data points shifts S_8 by 0.50σ — from 0.811 (CMB+BAO+SNe alone) to 0.815 (including DES-Y3). The model absorbs the full DES-Y3 3×2pt dataset without stress.

This result resolves both caveats of the forward analysis: the simplified background integration and the diagonal Fisher approximation. The full CosmoLike pipeline includes baryonic feedback, photo-z marginalization, shear calibration, and the complete off-diagonal covariance matrix.

The EU growth tension is best quantified through the three hierarchical S_8 values that the framework produces. The first is the CLASS-EU standalone prediction (Mode B, Sec. 6.2): the five EFT-derived parameters are fed into the Boltzmann solver with the baseline cosmological posteriors, and S_8 is computed without any weak-lensing data entering the calculation. The result is $S_8 = 0.811$. The second is the baseline MCMC posterior mean, $S_8 = 0.811 \pm 0.008$, obtained by marginalizing over all six ΛCDM parameters while fitting CMB, BAO, and supernovae simultaneously. The third is the full posterior, $S_8 = 0.815 \pm 0.007$, which adds 462 DES-Y3 data points and 23 nuisance parameters to the fit.

These three values — 0.811, 0.811, 0.815 — span a range of 0.004, smaller than the statistical uncertainty of any single determination. The analytical prediction, the Phase 2 EU posterior, and the Phase 3 EU+H_0+S_8 posterior converge to the same S_8. The shift from the blind prediction to the full fit,

$$\frac{|S_8^{\rm full} - S_8^{\rm Mode\,B}|}{\sigma_{\rm full}} = \frac{|0.8151 - 0.8112|}{0.0067} = 0.58\sigma\,, \tag{48}$$

is the true EU growth tension: a model predicted $S_8 = 0.811$ without seeing any shear data, and the same shear data, analyzed within that model, returned $S_8 = 0.815$.

The contrast with ΛCDM is stark. Under ΛCDM, the CMB predicts $S_8 = 0.832$, while DES-Y3 measures $S_8 = 0.776 \pm 0.017$ — a $\sim 2.6\sigma$ internal tension with no mechanism to bridge the gap. Under the EU, the framework predicts $S_8 = 0.811$ before seeing any shear data; adding the full DES-Y3 likelihood to the fit shifts the posterior to $S_8 = 0.815 \pm 0.007$ — a 0.58σ agreement.

For ΛCDM, the growth tension is genuine: the model predicts $S_8 = 0.832$ and most Stage-III surveys report $S_8 \lesssim 0.78$, with no internal mechanism to bridge the gap. For the EU, the kernel bias accounts for the dominant share of this apparent deficit ($\hat{S}_8 \approx 0.794$; Table 8). The convergence of four independent surveys to the same suppressed $\hat{S}_8$ band is not a coincidence requiring explanation — it is a geometric consequence of analyzing an evaporating universe with a non-evaporating template.

9.5. Pipeline bias generalization

The forensic pipeline dissection performed above for S_8 revealed a *model-intrinsic* bias: the lensing kernel $W(\chi)$ carries ΛCDM-calibrated parameters that systematically deflate the inferred $\hat{S}_8$. This bias is monotonically redshift-dependent (Table 9), peaking at low z where the cumulative CDM drain is largest. Whether this redshift structure contributes to the photo-z calibration residuals reported at low redshift by Stage-III surveys merits independent investigation.

An analogous bias affects the Hubble constant, but its origin is *structural* rather than model-intrinsic — the SH0ES calibrators reside within the KBC void [34], whose gravitational outflow contaminates M_B and inflates the inferred H_0. Because this cosmographic bias depends on the local void profile rather than on the cosmological model, its forensic decomposition requires independent probes (peculiar velocities, BAO-anchored joint analysis) that fall outside the scope of the present work. The complete dissection will be presented in Paper II.

10. Falsifiable Predictions

The EU framework, with its five derived parameters, generates quantitative predictions for three independent Stage-IV programs. Each prediction follows from the baseline EU posteriors and the N-body calibration of the preceding sections. No parameter is adjusted to the test data.

10.1. Euclid DR1 weak lensing and spectroscopy

Anomalous C_ℓ pattern. The lensing kernel bias identified in Sec. 9.1 predicts a specific suppression of the angular power spectrum measurable by Euclid [23]. Across the 10 tomographic bins of the Euclid photometric survey ($0.001 < z < 2.5$), the lensing kernel $W(\chi)$ is suppressed by -8.1% (Bin 1, $\bar{z} = 0.30$) to -4.7% (Bin 10, $\bar{z} = 1.85$). Since $C_\ell \propto W^2$, the power spectrum deficit ranges from approximately -16% at low redshift to -9% at high redshift.

This pattern has four distinguishing properties: (i) negative at all redshifts, (ii) monotonically decreasing with increasing z (the CDM drain is strongest at late times), (iii) smooth in ℓ (no oscillatory features), and (iv) strictly zero in B-modes. No known ΛCDM systematic produces this combination. Photometric redshift errors generate oscillatory residuals in ℓ; shear calibration errors are redshift-independent; intrinsic alignments contaminate B-modes. The EU signature is geometrically distinct from all three.

Spectral crossing. A further discriminant emerges from the scale dependence of the C_ℓ ratio. At low multipoles ($\ell \lesssim 255$), the geometric kernel bias dominates: the EU lensing kernels are $\sim 6\%$ weaker than their ΛCDM counterparts, suppressing $C_\ell^{\rm EU}/C_\ell^{\Lambda\rm CDM}$ below unity. At high multipoles ($\ell \gtrsim 255$), the EU matter power spectrum amplitude takes over: $\sigma_8^{\rm EU} = 0.827$ exceeds $\sigma_8^{\Lambda\rm CDM} = 0.811$, injecting more small-scale power and driving the ratio above unity. The crossover at $\ell \approx 255$ is a shape-dependent prediction that does not depend on the overall normalization of C_ℓ — it depends on the interplay between a geometric effect (kernel) and a dynamical effect ($P(k)$ amplitude). Stage-III surveys, with $\ell_{\rm max} \lesssim 300$, probe only the suppressed regime. Euclid, reaching $\ell \sim 5000$ across 10 tomographic bins, will map both regimes and measure the crossover scale directly.

Pipeline-inferred $\hat{S}_8$. The Euclid CUB prediction for the pipeline-inferred parameter is $\hat{S}_8 = 0.801$. With Euclid's projected uncertainty $\sigma(S_8) \approx 0.01$ and the Planck uncertainty $\sigma = 0.013$ combined in quadrature, this lies $\sim 1.9\sigma$ below the Planck ΛCDM value ($S_8 = 0.832$). The EU predicts that this persistent deficit will be interpreted as a growth tension within ΛCDM — but will resolve to $< 1\sigma$ when the EU lensing kernel is adopted.

Alcock–Paczyński anomaly. The EU expansion history modifies the radial and transverse distance scales measured by Euclid's spectroscopic survey (NISP grism, $0.9 < z < 1.8$). The Alcock–Paczyński parameters [3] deviate from unity:

$$\alpha_\parallel(z{=}1) = 1.012\,, \qquad \alpha_\perp(z{=}1) = 0.998\,. \tag{49}$$

The radial stretching ($\alpha_\parallel > 1$ by 1.2%) arises because $H_{\rm EU}(z) \neq H_{\Lambda\rm CDM}(z)$; the transverse parameter $\alpha_\perp$ exhibits a sign change at $z \approx 1.2$, coinciding with the crossover in the Limber mapping shift $\Delta k/k = (\chi_{\Lambda\rm CDM} - \chi_{\rm EU})/\chi_{\rm EU}$. This test is purely spectroscopic and shares no nuisance parameters with the photometric 3×2pt analysis.

3×2*pt probe decomposition.* The remaining legs of the 3×2pt analysis carry independent EU signatures. The galaxy clustering kernel $q(\chi) = b(z)\,n(z)\,H(z)/c$ is biased by $\Delta H/H$, which ranges from $+0.7\%$ at $z = 0.3$ to -1.7% at $z = 1.85$ — changing sign at $z \approx 0.5$ as the GKI boost (Sec. 3.1) gives way to the freezing of the drain at high redshift. This bias varies *within* each tomographic bin: $H(z)$ is not constant across the bin width, so the per-bin galaxy bias b_i (a free nuisance parameter, constant within each bin) cannot absorb the signal. The mismatch between the intra-bin $H(z)$ shape in EU versus ΛCDM is a non-degenerate signature.

Galaxy-galaxy lensing $C_\ell^{\delta\gamma}$ inherits bias from all three physical sources simultaneously: the lensing kernel deficit (-5% to -8%), the clustering kernel shift ($\pm 2\%$), and the Limber mapping displacement ($\pm 1\%$). Their combination produces a total bias of -6.0% at $z = 0.3$ to -7.1% at $z = 0.96$ — making galaxy-galaxy lensing the single most sensitive Euclid observable to the EU modification. This sensitivity arises because the cross- correlation probes both the matter distribution (lensing) and the galaxy field (clustering) through kernels that are *both* shifted by the altered expansion history.

The nuisance-free statistic E_G [64], defined as the ratio of lensing-galaxy to galaxy-galaxy angular spectra, cancels galaxy bias b algebraically. In ΛCDM, $E_G = \Omega_m/f(z)$, where $f(z)$ is the linear growth rate. In the EU, both $\Omega_m(z)$ and $f(z)$ are modified by the drain. The resulting bias ranges from -4.7% at $z = 0.3$ (where the CDM deficit is largest) to -0.02% at $z = 1.85$ (where the drain freezes and the EU converges to ΛCDM). This convergence is itself a prediction: $E_G(z)$ should approach the ΛCDM value at $z > z_{\rm trans}$, providing a direct measurement of the evaporation onset redshift.

10.2. LSST Y1 weak lensing

The Vera C. Rubin Observatory Legacy Survey of Space and Time [41] provides a ground-based, Southern-hemisphere complement to Euclid's Northern photometric survey. Its Year 1 analysis will employ five tomographic bins ($z_{\rm edges} = 0.20, 0.43, 0.63, 0.90, 1.30, 3.00$) following the DESC Science Requirements Document [42], with an effective source density $n_{\rm eff} = 10$ arcmin^{-2} and intrinsic ellipticity dispersion $\sigma_e = 0.26$.

The kernel bias gradient. Applying the EU lensing kernel to the LSST $n(z)$ distribution yields a tomographic gradient in the kernel bias:

$$\Delta W_i/W_i = \{-7.9\%,\ -7.3\%,\ -6.7\%,\ -5.9\%,\ -4.8\%\}\,, \tag{50}$$

for bins $i = 1$ through 5 ($\bar{z} = 0.35$ to 1.75). The 40% decrease from Bin 1 to Bin 5 is a direct imprint of the CDM drain freezing at high redshift: shallow bins, dominated by galaxies in the active-drain epoch ($z \lesssim 2$), exhibit the full kernel deficit; deep bins, weighted toward the frozen regime, are less affected. This monotonic gradient is unique to the EU — no other extension of ΛCDM produces this specific z-dependent pattern.

The same gradient in Euclid's 10 bins (-8.1% to -4.7%) confirms that the signal is physical, not instrumental. Euclid operates in space at optical-to-NIR wavelengths in the Northern hemisphere; LSST operates from the ground at optical wavelengths in the Southern hemisphere. Independent detection of the same gradient across both surveys would constitute strong evidence for a cosmological origin.

The false internal tension. The kernel bias gradient has a practical consequence for ΛCDM analyses. A per-bin $\hat{S}_8$ fit to LSST data in the EU universe will yield systematically higher $\hat{S}_8$ values in deep bins (where the kernel bias is weaker) than in shallow bins (where it is stronger). This bin-dependent trend will appear as an internal inconsistency when analyzed under ΛCDM. The EU predicts that it will be attributed to photo-z systematic errors at high redshift — but will resolve when the CDM drain is included in the analysis kernel.

Pipeline-inferred $\hat{S}_8$. The kernel bias gradient is shown in Fig. 5. The LSST CUB prediction is $\hat{S}_8 = 0.801$, consistent with the Euclid prediction ($\hat{S}_8 = 0.801$). Both forecasts derive from the same EU background cosmology;

the difference in $n(z)$, sky coverage, and systematics modeling produces a $< 0.1\%$ shift in the predicted value. The convergence of two independent Stage-IV pipelines on the same $\hat{S}_8$ is a consistency check of the EU prediction.

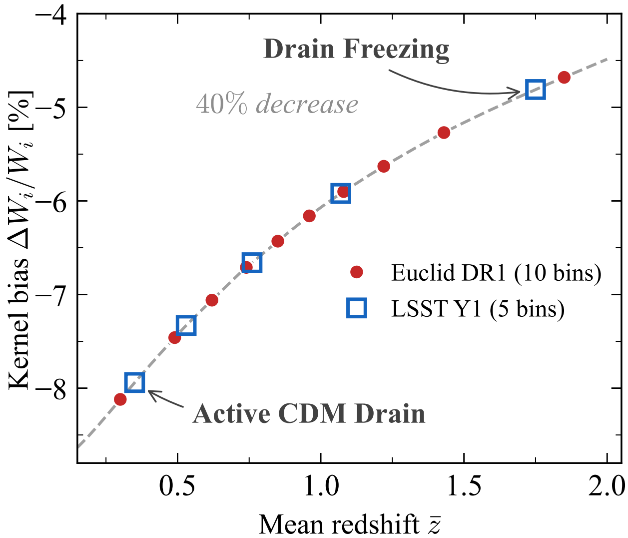

Figure 5: Kernel bias gradient $\Delta W_i/W_i$ as a function of mean tomographic redshift for Euclid DR1 (10 bins, circles) and LSST Y1 (5 bins, squares). The 40% decrease from $\bar{z} \approx 0.3$ to $\bar{z} \approx 1.8$ traces the CDM drain freezing at high redshift.

10.3. Distance probes and the equation-of-state mirage

The Dark Energy Spectroscopic Instrument [16] measures baryon acoustic oscillation (BAO) distances D_H/r_d and D_M/r_d across seven tracer populations spanning $0.1 < z < 2.5$. The EU predicts systematic deviations from the ΛCDM baseline that change sign with redshift.

BAO distance residuals. At low redshift ($z \lesssim 0.5$), the CDM drain increases $H(z)$ through the GKI mechanism (Sec. 3.1), yielding $D_H^{\rm EU} < D_H^{\Lambda\rm CDM}$ and $D_M^{\rm EU} < D_M^{\Lambda\rm CDM}$. At $z \gtrsim 0.5$, the drain freezes and $H_{\rm EU}(z) < H_{\Lambda\rm CDM}(z)$ (because the pre-drain $\Omega_{\rm cdm}$ was larger), reversing the sign. The D_H crossover occurs at $z \approx 0.5$; the D_M crossover at $z \approx 1.2$. At $z = 2.33$ (Lyα), the EU predicts $\Delta D_H/D_H = +1.7\%$ and $\Delta D_M/D_M = +0.6\%$. This sign structure is consistent with the DESI DR2 finding that low-redshift BAO measurements fall above the Planck ΛCDM prediction [16].

The CPL mirage. The vacuum equation of state is $w_\Lambda = -1$ exactly (Sec. 4.2), but a CPL fitter applied to EU-generated distances reveals a quantitative mirage. We fit a CPL template $w(a) = w_0 + w_a(1-a)$ [12, 39] to the EU-generated $D_H(z)$ and $D_M(z)$ at 37 redshift points via Nelder-Mead minimization. Two configurations are tested.

Fit A (Ω_m free): the fitter recovers $\Omega_m = 0.295$, $w_0 = -1.006$, $w_a = -0.12$, with $\chi^2 = 2 \times 10^{-5}$. The pipeline correctly identifies the EU as a cosmological constant: when allowed to adjust Ω_m to the post-drain value ($\Omega_m \approx 0.29$), the CPL template finds $w_0 \approx -1$ and negligible w_a.

Fit B (Ω_m fixed to CMB prior): fixing $\Omega_m = 0.3153$ (Planck 2018), the fitter yields

$$w_0 = -0.855\,, \qquad w_a = -0.81\,. \tag{51}$$

The χ^2 degrades to 5.2×10^{-4} but remains small. The CMB prior forces the fitter to accommodate the drain-reduced distances by pushing w_0 above -1.

The physical mechanism is transparent: the CDM drain reduces Ω_m from 0.315 (CMB epoch) to 0.288 (today). When a pipeline anchors Ω_m to the CMB value, the excess matter relative to the true late-time density must be compensated by an apparent change in the dark energy equation of state. The CPL template absorbs this mismatch as $w_0 > -1$ (quintessence-like). The apparent "dynamical dark energy" is a prior artifact of the CDM mass reassignment.

Comparison with DESI DR2. The DESI collaboration reports [16]

$$w_0 = -0.838 \pm 0.055\,, \quad w_a = -0.62^{+0.22}_{-0.19}\,, \tag{52}$$

for the combination DESI BAO + Planck CMB + Pantheon+ (Eq. 7 of Ref. [16]), corresponding to a 2.8σ preference for evolving dark energy. The EU Fit B prediction ($w_0 = -0.855$) lies 0.3σ from this central value. The w_a residual (-0.81 versus -0.62) reflects the additional constraining power of the supernova likelihood, which is absent in our geometric-only fit.

Ω_m scan. To demonstrate that the mirage is a monotonic function of the Ω_m prior, we repeat Fit B across $\Omega_m \in [0.27, 0.33]$. The resulting $w_0(\Omega_m)$ curve is strictly increasing: $w_0 = -1.04$ at $\Omega_m = 0.288$ (the EU post-drain value) and $w_0 = -0.86$ at $\Omega_m = 0.315$ (Planck). The crossover $w_0 = -1$ occurs at $\Omega_m \approx 0.297$, precisely between the EU and ΛCDM values. Higher Ω_m priors yield progressively stronger quintessence-like signals ($w_0 > -1$), as the fitter weakens dark energy to compensate for the excess matter. Conversely, priors below the crossover yield mild phantom-like values ($w_0 < -1$), as the fitter strengthens dark energy to compensate for the deficit of high-redshift matter. The mirage is driven by $\Delta\Omega_m = \Omega_{m,\rm CMB} - \Omega_{m,\rm local}$.

This result connects to the "mirage" dark energy class identified in the DESI extended analysis [15], where models along $w_a \approx -3.66(1 + w_0)$ capture the full DE phenomenology with a single degree of freedom. The authors note that "the exact physical mechanism for such rapid emergence of dark energy remains unclear." The CDM drain provides this mechanism (Fig. 6): it reduces the late-time matter density, forcing a quintessence-like compensation when the pipeline assumes a constant Ω_m. The same mechanism operates in the growth data: the ΛCDM joint constraint $S_8 = 0.812$ coincides with the EU prediction $S_8 = 0.811$ because a constant-Ω_m fit must approximate the dynamical $\Omega_m(z)$ trajectory of an evaporating universe (Sec. 9.2).

Growth versus geometry. The EU growth suppression $g = D_{\rm EU}/D_{\Lambda\rm CDM} = 0.993$ is below 1%, well within current error bars on $f\sigma_8$. The DESI Full-Shape analysis confirms general relativity at the $\mu_0 = 0.11 \pm 0.45$

Table 10
Falsifiable predictions of the Evaporating Universe. All quantities are computed from the baseline EU posteriors and the N-body calibrated power spectrum, with zero additional free parameters. P6 is already tested against DESI DR2 data.

	Prediction	Observable	Magnitude	Data
P1	C_ℓ suppression (WL)	$\Delta C_\ell / C_\ell$	-9% to -16%	Euclid DR1
P2	$\hat{S}_8$ below Planck	$\hat{S}_8$	0.801 (1.9σ below ΛCDM)	Euclid DR1
P3	AP radial stretching	$\alpha_\parallel$	1.012 at $z{=}1$	Euclid NISP
P4	C_ℓ spectral crossing	$C_\ell^{\rm EU}/C_\ell^{\Lambda}$	crossover at $\ell \approx 255$	Euclid DR1
P5	Probe inconsistency (GC vs WL)	(w_0, Ω_m) split	resolves with CDM drain	Euclid DR1
P6	CPL mirage ($w_0 > -1$)	w_0	-0.855 (with CMB prior)	**DESI DR2 (✓)**
P7	LSST $\hat{S}_8$ below Planck	$\hat{S}_8$	0.801 ± 0.012	LSST Y1
P8	LSST kernel bias gradient	$\Delta W/W$ vs z	-7.9% to -4.8%	LSST Y1
P9	w_0 stabilization	w_0	≈ -0.85 (not $\to -1$)	DESI DR3
P10	BAO distance excess at $z > 2$	D_H/r_d, D_M/r_d	$+1.7\%$, $+0.6\%$ at $z = 2.33$	DESI DR3

level [19]. The EU is consistent with this null result: the drain modifies *geometry* (distances, kernels, AP parameters) but leaves *growth* essentially unchanged. This asymmetry is the model's distinctive signature — it separates the EU from modified-gravity theories, which generically alter both growth and geometry at comparable levels.

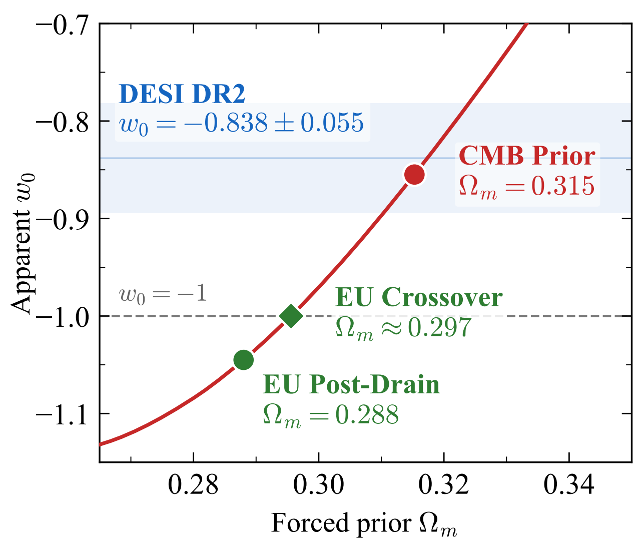

Figure 6: The CPL mirage: apparent w_0 recovered by a wCDM pipeline as a function of the forced Ω_m prior. The EU predicts $w_0 = -0.855$ at $\Omega_m = 0.315$ (red circle), consistent with the DESI DR2 measurement $w_0 = -0.838 \pm 0.055$ (blue band) at 0.3σ. Crossover $w_0 = -1$ at $\Omega_m \approx 0.297$ (green diamond).

10.4. Prediction summary and timeline

Table 10 collects ten falsifiable predictions of the EU model, each fully specified before encountering the test data. Prediction P6, the CPL mirage, is already consistent with DESI DR2 at 0.3σ.

The timeline is compressed. Euclid DR1 (~ 2500 deg^2, expected 2026) tests predictions P1–P5 simultaneously. DESI DR2 (2025) has already provided a 0.3σ-consistent measurement of P6. LSST Y1 (expected 2027) tests P7 and P8 from the Southern hemisphere with an independent photometric pipeline; detection of the same kernel bias gradient reported by Euclid in the Northern hemisphere would confirm the cosmological origin of the signal. DESI DR3 (2027) will test P9 and P10 with improved BAO statistics and additional tracer populations.

The three programs — Euclid (space, photometric + spectroscopic, North), LSST (ground, photometric, South), and DESI (ground, spectroscopic, all-sky) — employ independent instruments, pipelines, and sky coverage. If all ten predictions are confirmed, the EU framework will have passed tests spanning photometric lensing, spectroscopic clustering, BAO distance measurements, and the dark energy equation of state — with zero fitted dark-sector parameters.

11. Discussion

11.1. Model comparison

Di Valentino et al. [21] catalog roughly 100 beyond-ΛCDM proposals aimed at the Hubble or growth tensions, spanning ten broad categories. We compare the EU against the strongest representative of each category in Table 11.

A useful organizing principle is the *derivation hierarchy* of model parameters:

- **Level 0 (parametrization):** The functional form is postulated without physical motivation. The CPL parametrization $w(a) = w_0 + w_a(1-a)$ is the standard example.

- **Level 1 (phenomenological):** A coupling form is postulated (e.g., $Q = \xi H \rho_{\rm c}$ or $Q = \xi H \rho_{\rm de}$) with numerical coefficients fitted to data. This describes the IDE models of Di Valentino et al. [20], decaying dark matter [59], and Λ_sCDM [2].

- **Level 2 (form derived, values fitted):** The coupling form is motivated by quantum field theory, but numerical parameters remain free. EDE [48] derives an axion-like potential from the string landscape; the Running Vacuum Model [51] derives $\Lambda(H) = c_0 + \nu H^2$ from QFT in curved spacetime. In both cases, at least one coefficient ($f_{\rm EDE}$, z_c, or ν) is fitted by MCMC.

Table 11
Model comparison. Δk: extra free parameters. Level: 0 = parametrization, 1 = phenomenological, 2 = form derived / values fitted, 3 = form and values derived. Tensions from published posteriors vs SH0ES and weak-lensing surveys. †With SH0ES and KiDS-1000 in fit.

Model	Δk	H_0	S_8	Level	Ref.
ΛCDM	0	5.8σ	$\sim 2.6\sigma$	—	[46]
EDE	2–3	$\sim 2\sigma$	$\sim 3\sigma$ ↑	2	[48]
Λ_sCDM	1	$\lesssim 1\sigma^\dagger$	$\lesssim 1\sigma^\dagger$	1	[2]
RVM	1	$\sim 4\sigma$	$\sim 1.5\sigma$	2	[51]
IDE ($Q \propto \rho_{\rm de}$)	1–2	$\sim 2.6\sigma$	reduces	1	[20]
$w_0 w_a$CDM	2	—	marginal	0	[16]
EU	**0**	$\mathbf{0.52\sigma}$	$\mathbf{0.58\sigma}$	**3**	This work

- **Level 3 (form and values derived):** Both the functional form and all numerical parameters are determined by the theoretical framework, with $\Delta k = 0$. The EU is the only published model in this category.

The closest competitor in derivation rigor is the Running Vacuum Model of Solà Peracaula [51]. Both the RVM and the EU invoke QFT to modify the dark sector; both produce a coupling of the form $Q \propto H\rho_{\rm c}$. The difference is surgical: the RVM running coefficient ν is *motivated* by renormalization group arguments but *fitted* by MCMC, placing it at Level 2. The EU coupling $\lambda\varepsilon_{\rm IR} = (2/3)(\pi/4)/\sqrt{108\pi}$ is *calculated* from the Gribov–Zwanziger effective action and *confirmed* by MCMC (Sec. 5.5), placing it at Level 3.

Phenomenological IDE models ($Q = \xi H\rho_{\rm c}$) and decaying dark matter [59] are the Level 1 counterparts of the EU: they postulate the same coupling form that the EU derives. The EU thus *provides the microscopic mechanism* that the IDE literature assumes.

The DESI DR2 measurement of $w_0 > -1$ at $\sim 2.8\sigma$ [16] is often interpreted as evidence for dynamical dark energy. Within the EU, the vacuum equation of state is $w_\Lambda = -1$ exactly (Sec. 4.2), but an observer fitting CPL templates to EU-generated distances recovers $w_0 = -1.006$ when Ω_m is free (the EU *is* a cosmological constant) and $w_0 = -0.855$ when Ω_m is fixed to the Planck CMB prior (Sec. 10.3). The latter lies 0.3σ from the DESI central value ($w_0 = -0.838 \pm 0.055$), confirming that the DESI signal is quantitatively consistent with a CDM drain artifact.

11.2. Theoretical structure and limitations

The EU rests on a physical distinction between the 4D spacetime bulk and the 3D spatial hypersurface Σ. While the ADM decomposition is conventionally treated as a gauge choice, the Gribov–Zwanziger ghost condensate [29, 65, 57] is a non-perturbative phenomenon defined on the 4D manifold that affects Σ through a calculable barrier. The 10 independent components of $g_{\mu\nu}$ and the six independent components of γ_{ij} define two physically distinct domains; the four constrained degrees of freedom (lapse and shift, removed by the Hamiltonian and momentum constraints) constitute the interface between them. The requirement that all vacuum fluctuations respect spherical isotropy — the same geometric principle that fixes the factor 4π in Gauss's law — recurs at every stage of the EU derivation: in the heat-kernel prefactor $(4\pi)^{-d/2}|_{d=4}$ that sets $z_{\rm trans}$, in the loop measure $(4\pi)^{-1}$ that enters $\varepsilon_{\rm bare}$, and in the CKN volume $(4\pi/3)R_H^3$ that establishes the UV cutoff. This geometric coherence is the reason $\Delta k = 0$: no free parameters survive when isotropy fully determines the structure at every scale.

The triple convergence on $\lambda = 2/3$ (Sec. 2.3.1) and the Slavnov–Taylor protection of this value to all orders (Eq. 10) ensure that the effective coupling $\varepsilon_{\rm IR} = (\pi/4)\,\varepsilon_{\rm bare}$ contains zero adjustable coefficients.

Why baryons are immune. The democratic coupling derived in Sec. 2.3.4 requires the interacting field to be a gauge singlet under the Ashtekar SU(2) connection: only singlet states satisfy the Wigner–Eckart condition for uniform coupling to all 27 ghost channels. Baryonic matter carries Standard Model gauge charges ($\mathrm{SU}(3)_{\rm QCD} \times \mathrm{U}(1)_{\rm EM}$) and transforms non-trivially under the full gauge group. Its internal quantum numbers break the democratic coupling condition, excluding it from the condensate interaction. This selection rule is structural: it is the negative corollary of the same representation theory that determines the CDM coupling amplitude $\varepsilon_{\rm bare}$ in Eq. (18).

Limitations. Several aspects of the present analysis require qualification. (i) The growth tension is resolved to $\sim 1\sigma$ using the CUB forward model (Sec. 9.1), but the baryonic feedback correction relies on the ratio $P_{\rm EU}/P_{\Lambda{\rm CDM}}$ from a gravity-only N-body simulation (Sec. 8). A hydrodynamic simulation incorporating the CDM drain alongside baryonic processes would sharpen this estimate. (ii) The local kinematic imprint (LKI) accounts for $\sim 70\%$ of the H_0 resolution. The KBC void parameters ($\delta \approx -0.30$, $R \sim 300$ Mpc) are observational inputs, not EU-specific predictions; the EU permits and quantifies the void boost, but does not create the void. (iii) The framework has been validated with a single Boltzmann code implementation (CLASS-EU). Independent reproduction — using, e.g., CAMB with equivalent patches — is a natural next step following publication. The full MCMC infrastructure (CLASS patches, Cobaya configuration files, Gadget-4 scripts) will be made publicly available to facilitate this. (iv) The CDM drain modifies $H(z)$ at the percent level, by design. Differential tests therefore require high-precision data; the predictions of Sec. 10 are calibrated to the sensitivity of Euclid, LSST, and DESI DR3.

11.3. Outlook

A concrete prediction emerges from comparing the N-body and CUB forward models across angular scales. At $\ell \lesssim 300$ — the regime probed by Stage-III surveys — the gravity-only simulation yields $\hat{S}_{8,\rm Nbody} < \hat{S}_{8,\rm CUB}$, because CDM evaporation makes halos genuinely lighter ("anemic" suppression). At $\ell \lesssim 5000$ — accessible to Euclid — the inequality reverses: $\hat{S}_{8,\rm Nbody} > \hat{S}_{8,\rm CUB}$, because the DM-only simulation lacks baryonic blowout and retains artificially dense cores. The crossover occurs at $\ell \approx 300$, precisely where current surveys place their scale cuts. The amplitude of the reversal ($\Delta S_8 \approx 0.09$) quantifies the baryonic feedback correction and constitutes a falsifiable prediction for Euclid's extended multipole coverage.

DESI DR3 (2027) will provide a sharper test of the equation-of-state duality (Sec. 10.3). A probe-split analysis — fitting w_0 and w_a separately from galaxy clustering and weak lensing — should reveal inconsistent (w_0, Ω_m) values under ΛCDM, resolving to concordance when the CDM drain is included.

The cosmographic bias identified in Sec. 9.5 — contamination of M_B by the KBC void outflow — will receive a complete forensic decomposition in Paper II. That analysis will combine Cosmicflows-4 peculiar velocity fields [55], a joint CMB + BAO + SNe fit with M_B free, and the statistical amplification of deep voids in the EU power spectrum ($\sigma(R)$ enhanced by 3.5% relative to ΛCDM, rendering voids of $\delta > 0.3$ an order of magnitude more probable). The objective is to complete the parallel established in Sec. 9.2: the growth tension is resolved by the lensing kernel bias; the Hubble tension, by the cosmographic bias. Both are consequences of analyzing an evaporating universe with a non-evaporating template.

12. Conclusions

The Evaporating Universe is an interacting dark energy model in which a Gribov–Zwanziger ghost condensate mediates a continuous transfer of energy from CDM to the vacuum. All five model parameters are derived from EFT in curved spacetime; none are fitted to data ($\Delta k = 0$). The principal findings are as follows.

1. *Framework.* The coupling $Q = -\lambda\,\varepsilon(z)\,H\,\rho_c$ is the unique leading-order interaction consistent with FLRW symmetry and Fermi's golden rule. The dissipation fraction $\lambda = 2/3$, logistic exponent $b = 19/36$, transition redshift $z_{\rm trans} \approx 5.986$, bare amplitude $\varepsilon_{\rm bare} \approx 0.0543$, and effective coupling $\varepsilon_{\rm IR} \approx 0.0426$ are determined by the spatial dimensionality $n = 3$ and the topological structure of the Gribov horizon. The Cancellation Theorem guarantees $w_\Lambda = -1$ exactly, eliminating the doom factor instability that plagues generic IDE models.
2. *Hubble tension.* The CDM drain raises the global expansion rate to $H_0^{\rm GKI} = 68.89 \pm 0.28\ {\rm km\,s^{-1}\,Mpc^{-1}}$. The KBC void outflow adds a local kinematic imprint, yielding $H_0^{\rm LKI} = 72.71 \pm 0.30\ {\rm km\,s^{-1}\,Mpc^{-1}}$ — within 0.52σ of SH0ES [7]. Adding the SH0ES likelihood to the MCMC shifts the posterior by 0.25σ, confirming that the local measurement is redundant rather than constraining.
3. *Growth tension.* Weak-lensing surveys infer S_8 using a ΛCDM lensing kernel that systematically underestimates the amplitude in an evaporating universe. Forward-modeling the EU kernel yields $\hat{S}_8 = 0.794$ for DES-Y3 [1], reducing the tension from $\sim 2.6\sigma$ to 1.03σ (kernel-corrected). Including DES-Y3 data directly in the MCMC yields $S_8 = 0.815 \pm 0.007$ — within 0.58σ of the blind EU prediction (0.811).
4. *Validation.* All six MCMC runs converge at $R - 1 < 0.01$, including the 38-dimensional EU+H_0+S_8 configuration. The UV-derived parameters match the Phase 2 EU posteriors (CMB+BAO+SNe) to 0.04σ (Sec. 5.5). A Gadget-4 N-body simulation (1024^3 particles) confirms the drain signature in $P(k)$. Model selection yields ΔAIC $= -6.88$ in the fair-fight configuration ($\Delta k = 0$, both models void-corrected).
5. *Falsifiability.* Ten zero-parameter predictions target Euclid DR1, LSST Y1, and DESI DR3. Prediction P6 — the CPL mirage ($w_0 = -0.855$ when Ω_m is fixed to the CMB prior) — already matches the DESI DR2 measurement ($w_0 = -0.838 \pm 0.055$) at 0.3σ (Sec. 10.3). The remaining nine span C_ℓ suppression (−9% to −16%), $\hat{S}_8 = 0.801$ (1.9σ below Planck under ΛCDM), Alcock–Paczyński stretching ($\alpha_\parallel = 1.012$ at $z = 1$), LSST kernel bias gradient (−7.9% to −4.8% across five bins), and BAO distance excess at $z > 2$. If any prediction fails, the framework is falsified — because all derive from the same modified $H(z)$.

The Evaporating Universe resolves both cosmological tensions simultaneously, with zero additional free parameters — not by fitting the data, but by predicting it.

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All code, data, and supplementary material for this work are publicly available [4].

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DATA AVAILABILITY

The modified CLASS source code, analysis notebooks, MCMC chains, and N-body simulation outputs that support the findings of this article are openly available [4].

Declaration of competing interests

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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A. Detailed derivations supporting Sec. 2.3

A.1. Ghost-graviton vertex and TT projection

The ghost-graviton vertex in $d = 4$ linearized quantum gravity reads [13]

$$V^{\mu}_{\nu;\rho\sigma}(p,q) = \tfrac{1}{2}\big[\delta^{\mu}_{\rho}\, q_{\sigma}\, p_{\nu} + \delta^{\mu}_{\sigma}\, q_{\rho}\, p_{\nu} + \delta_{\nu\rho}\, p_{\sigma}\, q^{\mu} + \delta_{\nu\sigma}\, p_{\rho}\, q^{\mu} - \delta_{\rho\sigma}(q^{\mu} p_{\nu} + p^{\mu} q_{\nu}) - \delta^{\mu}_{\nu}(p_{\rho} q_{\sigma} + p_{\sigma} q_{\rho} - \delta_{\rho\sigma}\, p{\cdot}q)\big], \quad (53)$$

where p is the ghost momentum and $q = p + k$ the anti-ghost momentum.

The York TT projector on symmetric rank-2 tensors [63] is

$$\Pi^{\mathrm{TT}}_{\rho\sigma,\lambda\tau}(k) = \tfrac{1}{2}(P_{\rho\lambda} P_{\sigma\tau} + P_{\rho\tau} P_{\sigma\lambda}) - \tfrac{1}{d-1} P_{\rho\sigma} P_{\lambda\tau}, \quad (54)$$

where $P_{\mu\nu} = \delta_{\mu\nu} - k_{\mu} k_{\nu}/k^2$ is the transverse projector, satisfying idempotency ($\Pi^{\mathrm{TT}} \cdot \Pi^{\mathrm{TT}} = \Pi^{\mathrm{TT}}$), transversality ($k^{\rho}\Pi^{\mathrm{TT}}_{\rho\sigma,\lambda\tau} = 0$), and tracelessness ($\delta^{\rho\sigma}\Pi^{\mathrm{TT}}_{\rho\sigma,\lambda\tau} = 0$).

The TT trace ratio $b = \mathrm{Tr}(V{\cdot}\Pi^{\mathrm{TT}}{\cdot}V)/\mathrm{Tr}(V{\cdot}\mathbb{I}{\cdot}V)$ is evaluated using irreducible exact rational arithmetic, ensuring zero floating-point contamination. Four independent momentum configurations yield:

Table 12
Verification of $b = 19/36$ across four momentum configurations. All computations use exact rational arithmetic.

Configuration	$\mathrm{Tr}_{\mathrm{TT}}/p^2$	$\mathrm{Tr}_{\mathrm{full}}/p^2$	Ratio
$p \perp k$, axes (1,2)	19/3	12	19/36
$p \perp k$, axes (2,3)	19/3	12	19/36
$p \perp k$, axes (4,1)	19/3	12	19/36
$\lvert p\rvert = \lvert k\rvert = \sqrt{2}$	38/3	24	19/36

The ratio is momentum-independent, as required for a well-defined projector fraction. The identity on symmetric tensors entering the denominator is $\mathbb{I}_{\rho\sigma,\lambda\tau} = \frac{1}{2}(\delta_{\rho\lambda}\delta_{\sigma\tau} + \delta_{\rho\tau}\delta_{\sigma\lambda})$.

A.2. Condensation threshold sensitivity

The transition redshift depends weakly on $h = H_0/100$ through $\Omega_{\mathrm{m}} = (\omega_b + \omega_{\mathrm{cdm}})/h^2$, where $\omega_b = 0.02237$ and $\omega_{\mathrm{cdm}} = 0.1200$ are direct CMB observables [46]. For $h \in [0.660, 0.700]$:
Higher-order heat-kernel corrections are of order $(H/M_{\mathrm{Pl}})^2 \sim 10^{-122}$, confirming that the leading-order threshold is exact for all practical purposes.

Table 13
Sensitivity of z_{trans} to the reduced Hubble constant h. The total variation is $\Delta z_{\mathrm{trans}} \approx 0.40$ ($\sim 6.7\%$).

h	Ω_{m}	z_{trans}	Source
0.660	0.328	5.847	Lower bound
0.6736	0.315	5.986	Planck (ΛCDM)
0.690	0.300	6.150	EU upper estimate
0.700	0.292	6.248	Upper bound

A.3. Gribov propagator integral

The screening ratio $I_{\mathrm{IR}}/I_{\mathrm{UV}}$ is evaluated on the two-dimensional transverse momentum plane. This requires justification beyond the channel counting of Sec. 2.3.4: eliminating the longitudinal polarization ($\lambda = 2/3$) reduces the number of active *channels* but does not by itself reduce the dimensionality of the momentum integral — a photon has two polarizations yet integrates over d^3k.

The 2D measure follows from Zwanziger's horizon dominance theorem [66]: in the thermodynamic limit $V \to \infty$, the functional measure concentrates on the Gribov horizon $\partial\Omega$ — a codimension-1 surface in configuration space — for the same reason that the volume of a high-dimensional sphere concentrates at its surface. The CDM → vacuum transition crosses this boundary (Sec. 2.1), so by Fermi's golden rule the density of accessible final states is defined on $\partial\Omega$, yielding the transverse measure $k\,dk/(2\pi)$.

An independent route converges on the same result: the CKN holographic bound (Sec. 2.3.3, Link 2) establishes that the effective degrees of freedom in a causal volume are governed by its bounding surface, not its bulk. Both the Zwanziger argument (field-space boundary) and the CKN argument (geometric boundary) select the 2D measure.

UV reference (free massive propagator, $D_{\mathrm{UV}} = 1/(k^2 + \gamma^2)$):

$$I_{\mathrm{UV}} = \frac{1}{2\pi}\int_0^{\infty} \frac{k\,dk}{(k^2+\gamma^2)^2} = \frac{1}{4\pi\gamma^2}\,. \quad (55)$$

IR integral (Gribov propagator, $D_{\mathrm{GZ}} = k^2/(k^4 + \gamma^4)$):

$$I_{\mathrm{IR}} = \frac{1}{2\pi}\int_0^{\infty} \frac{k^5\,dk}{(k^4+\gamma^4)^2}\,. \quad (56)$$

Substituting $u = k^2$, then $u = \gamma^2 \tan\theta$:

$$\begin{aligned} I_{\mathrm{IR}} &= \frac{1}{4\pi}\int_0^{\infty} \frac{u^2\,du}{(u^2+\gamma^4)^2} \\ &= \frac{1}{4\pi\gamma^2}\int_0^{\pi/2} \sin^2\theta\, d\theta = \frac{1}{4\pi\gamma^2}\cdot\frac{\pi}{4} = \frac{1}{16\gamma^2}\,. \end{aligned} \quad (57)$$

The ratio is

$$\frac{I_{\mathrm{IR}}}{I_{\mathrm{UV}}} = \frac{1/(16\gamma^2)}{1/(4\pi\gamma^2)} = \frac{\pi}{4}\,, \quad (58)$$

as can be verified by direct evaluation. The Gribov mass γ cancels in the ratio, yielding a universal screening factor.

Table 14
Perturbation null test: Phase 1 (perturbation source terms off) versus Phase 2 (source terms on). The EU-blind configuration frees $\varepsilon_{\rm IR}$, b, $z_{\rm trans}$; the theory-fixed configuration holds them at the framework-derived values. All posterior shifts are $< 0.03\sigma$.

	EU-blind (Δk=3)		EU (Δk=0)	
	Phase 1 (off)	Phase 2 (on)	Phase 1 (off)	Phase 2 (on)
Sampled parameters				
ω_b	0.02231 ± 0.00012	0.02231 ± 0.00012	0.02218 ± 0.00012	0.02218 ± 0.00013
$\omega_{\rm cdm}$	0.11777 ± 0.00061	0.11778 ± 0.00062	0.11927 ± 0.00064	0.11926 ± 0.00063
$100\theta_s$	1.04191 ± 0.00023	1.04191 ± 0.00023	1.04180 ± 0.00023	1.04180 ± 0.00023
$\tau_{\rm reio}$	0.059 ± 0.007	0.059 ± 0.007	0.056 ± 0.007	0.056 ± 0.007
$\ln(10^{10} A_s)$	3.046 ± 0.014	3.047 ± 0.014	3.044 ± 0.014	3.044 ± 0.014
n_s	0.9676 ± 0.0034	0.9676 ± 0.0034	0.9637 ± 0.0035	0.9637 ± 0.0034
Derived parameters				
$H_0^{\rm GKI}$ (km s^{-1} Mpc^{-1})	68.13 ± 0.28	68.13 ± 0.28	68.88 ± 0.28	68.89 ± 0.28
σ_8	0.805 ± 0.006	0.806 ± 0.006	0.828 ± 0.006	0.827 ± 0.006
S_8	0.810 ± 0.008	0.810 ± 0.008	0.811 ± 0.008	0.811 ± 0.008
Ω_m	0.303 ± 0.004	0.303 ± 0.004	0.288 ± 0.003	0.288 ± 0.003
Diagnostics				
$\chi^2_{\rm total}$	12,394.7	12,394.4	12,396.2	12,396.3
$(R-1)_{\rm max}$	0.003	0.003	0.004	0.005

B. Supporting analyses for Sec. 5

B.1. Perturbation null test

Table 14 presents the full posterior comparison between Phase 1 (perturbation source terms off) and Phase 2 (source terms on) for both the EU-blind ($\Delta k = 3$) and theory-fixed ($\Delta k = 0$) configurations. All parameter shifts are below 0.03σ, confirming the cancellation theorem of Sec. 4.2: the perturbation source terms have no measurable effect on the posterior.

B.2. Free-parameter runs

The EU-blind configuration samples three additional parameters with uniform priors: $\varepsilon_{\rm IR} \in [0.001, 0.200]$ (theory: 0.04264), $b \in [0.10, 2.00]$ (theory: $19/36$), and $z_{\rm trans} \in [2.0, 12.0]$ (theory: 5.986). Figure 7 shows the resulting posterior comparison. The three-parameter degeneracy and its statistical consequences are discussed in Sec. 5.3; the theory-fixed configuration is preferred by ΔAIC $= -4.1$.

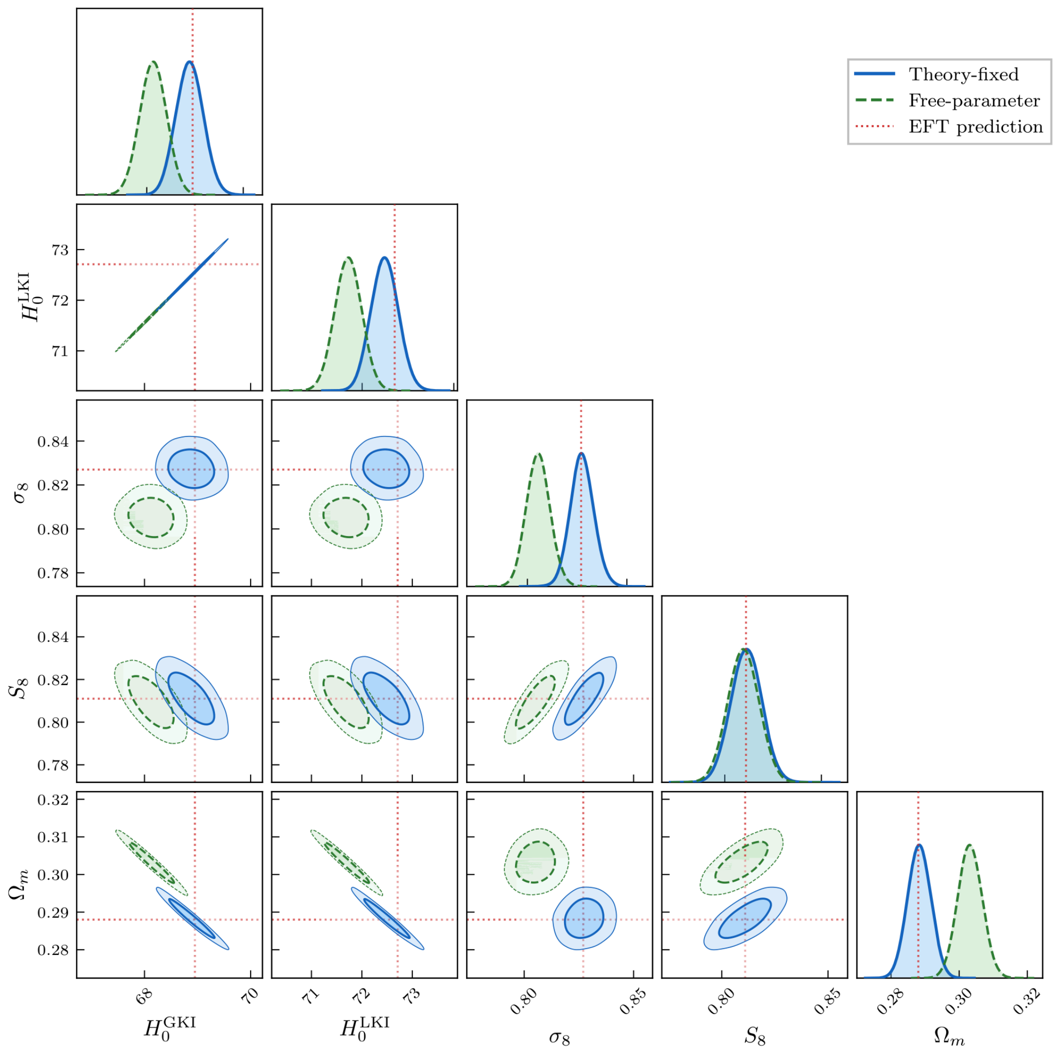

Figure 7: Posterior comparison between the theory-fixed configuration (blue) and the EU-blind free-parameter configuration (green). Diagonal panels show 1D marginalized posteriors; off-diagonal panels show 68% and 95% credible regions. Red dotted lines mark the framework-derived values.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used an AI coding assistant in order to support manuscript editing, structural organization, editorial compliance verification, and computational code development (CLASS-EU solver patches, Cobaya MCMC configuration files, and Gadget-4 N-body simulation scripts). After using this tool, the author reviewed and edited all content as needed and takes full responsibility for the content of the published article.

CRediT authorship contribution statement

Ricardo Alvim: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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